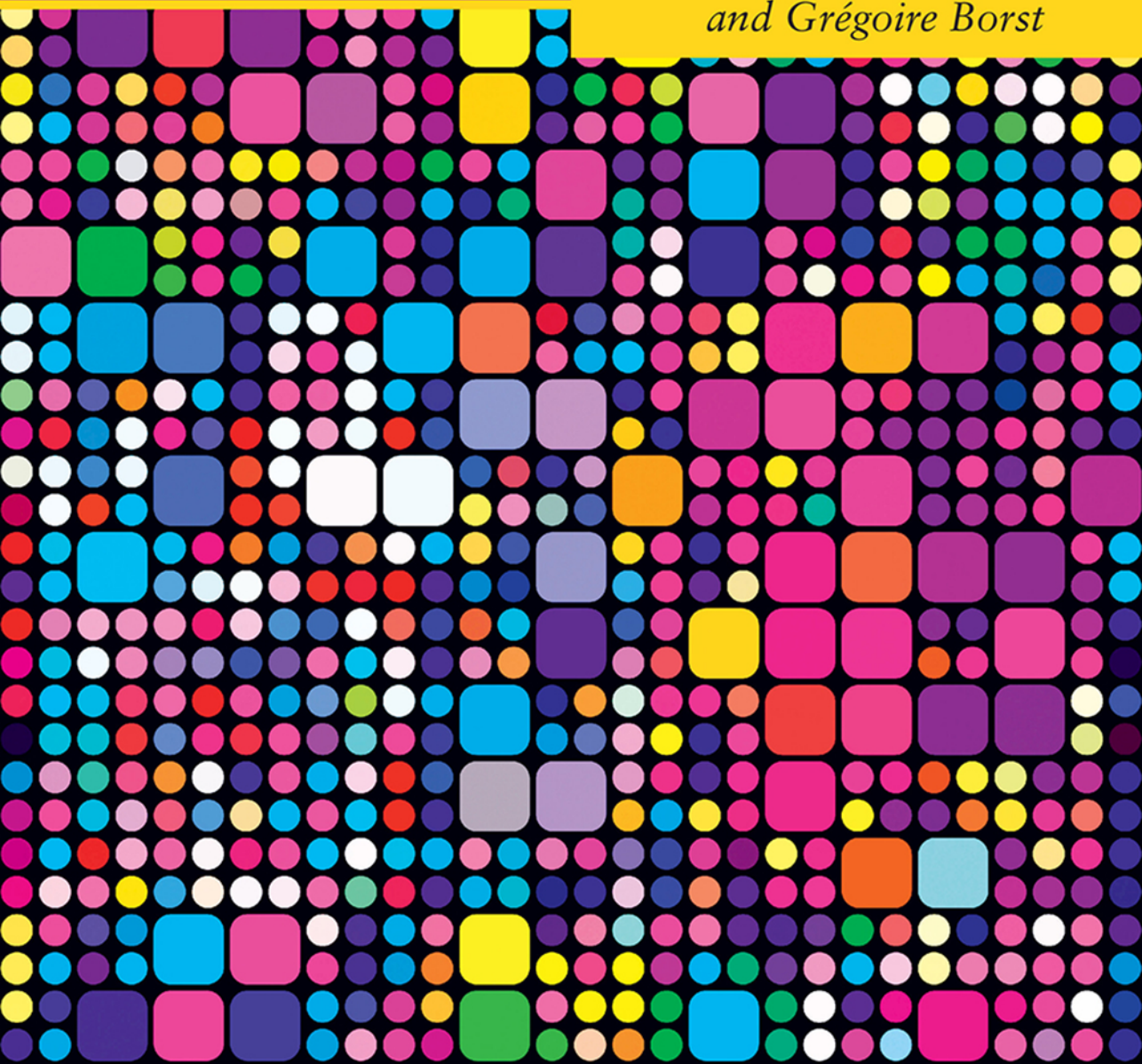
THE CAMBRIDGE HANDBOOK OF COGNITIVE DEVELOPMENT

Edited by Olivier Houdé and Grégoire Borst



The Cambridge Handbook of Cognitive Development

How does cognition develop in infants, children and adolescents? This handbook presents a cutting-edge overview of the field of cognitive development, spanning basic methodology, key domain-based findings and applications. Part I covers the neurobiological constraints and laws of brain development, while Part II covers the fundamentals of cognitive development from birth to adulthood: object, number, categorization, reasoning, decision-making and socioemotional cognition. Part III covers educational and school-learning domains, including numeracy, literacy, scientific reasoning skills, working memory and executive skills, metacognition, curiosity-driven active learning and more. Featuring chapters written by the world's leading scholars in experimental and developmental psychology, as well as in basic neurobiology, cognitive neuroscience, computational modelling and developmental robotics, this collection is the most comprehensive reference work to date on cognitive development of the twenty-first century. It will be a vital resource for scholars and graduate students in developmental psychology, neuroeducation and the cognitive sciences.

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The Cambridge Handbook of Cognitive Development

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Contents

List of Figures *page* ix
List of Tables xiii
List of Contributors xiv

Introduction: Cognitive Development Studies: From the History of Psychology to the Current Trends in Cognitive Sciences
OLIVIER HOUDÉ AND GRÉGOIRE BORST 1

Part I Neurobiological Constraints and Laws of Cognitive Development

1 How Life Regulation and Feelings Motivate the Cultural Mind: A Neurobiological Account
ANTONIO DAMASIO AND HANNA DAMASIO 15

2 Epigenesis, Synapse Selection, Cultural Imprints, and Human Brain Development: From Molecules to Cognition
JEAN-PIERRE CHANGEUX 27

3 Mapping the Human Brain from the Prenatal Period to Infancy Using 3D Magnetic Resonance Imaging: Cortical Folding and Early Grey and White Maturation Processes
ARNAUD CACHIA, JEAN-FRANÇOIS MANGIN, AND JESSICA DUBOIS 50

4 Development and Maturation of the Human Brain, from Infancy to Adolescence
TOMÁŠ PAUS 85

5 Genetic and Experiential Factors in Brain Development: The Examples of Executive Attention and Self-regulation
MICHAEL I. POSNER AND MARY K. ROTHBART 105

6 The Brain Basis Underlying the Transition from Adolescence to Adulthood
BEATRIZ LUNA, ORMA RAVINDRANATH, BART LARSEN, AND ASHLEY PARR 122

Part II Fundamentals of Cognitive Development from Infancy to Adolescence and Young Adulthood

Introduction: Assembling the Building Blocks of Cognition in a Non-linear Dynamical System of Development
GRÉGOIRE BORST AND OLIVIER HOUDÉ 141

Subpart II.1 Infancy: The Roots of Human Thinking

7 Differences between Humans, Great Apes and Monkeys in Cognition, Communication, Language and Morality
JACQUES VAUCLAIR 151

8 Infants' Physical Reasoning and the Cognitive Architecture that Supports It
YI LIN, MAAYAN STAVANS, AND RENÉE BAILLARGEON 168

9 Infant Categorization
DAVID H. RAKISON, DEON T. BENTON, AND PHUONG NGOC DINH 195

10 Foundational Considerations: Does Primitive Number Sense Provide a Foothold for Learning Formal Mathematics?
KRISTY VANMARLE 216

11 How Sophisticated Is Infants' Theory of Mind?
ROSE M. SCOTT, ERIN ROBY, AND RENÉE BAILLARGEON 242

12 Social Cognition and Moral Evaluation in Early Human Childhood
ANDREAS FALCK, BRENT STRICKLAND, AND PIERRE JACOB 269

13 Scientific Thinking and Reasoning in Infants and Young Children
MARIEL K. GODDU AND ALISON GOPNIK 299

14 Computational Approaches to Cognitive Development: Bayesian and Artificial-Neural-Network Models
ARDAVAN S. NOBANDEGANI AND THOMAS R. SHULTZ 318

Subpart II.2 Childhood and Adolescence: The Development of Human Thinking

15 Development of Qualitative Thinking: Language and Categorization
SUSAN A. GELMAN 341

16 Development of Numerical Knowledge
ROBERT S. SIEGLER 361

17 Numerical Cognition and Executive Functions: Development As Progressive Inhibitory Control of Misleading Visuospatial Dimensions
MARGOT ROELL, OLIVIER HOUDÉ, GRÉGOIRE BORST, AND ARNAUD VIAROUGE 383

18 Developing Theory of Mind and Counterfactual Reasoning in Children
JOSEF PERNER, EUGENIA KULAKOVA, AND EVA RAFETSEDER 408

19 Development of Executive Function Skills in Childhood: Relevance for Important Life Outcomes
KEITH HAPPANEY AND PHILIP DAVID ZELAZO 427

20 Developing Cognitive Control and Flexible Adaptation during Childhood
NICOLAS CHEVALIER AND AGNÈS BLAYE 452

21 Reasoning Bias and Dual Process Theory: Developmental Considerations and Current Directions
DARREN FREY AND WIM DE NEYS 472

22 Social Cognitive Development: The Intergroup Context
LISA CHALIK, ANTONIA MISCH, AND YARROW DUNHAM 481

23 Behavioral and Neural Development of Cognitive Control and Risky Decision-Making across Adolescence
NEELTJE E. BLANKENSTEIN, JISKA S. PEPER, AND EVELINE A. CRONE 500

24 The Triadic Neural Systems Model through a Machine-Learning Mill
MONIQUE ERNST, JOSH GOWIN, AND CLAUDIE GAILLARD 516

Part III Education and School-Learning Domains

25 Linking Cognitive Neuroscientific Research to Educational Practice in the Classroom
NATHAN T. T. LAU AND DANIEL ANSARI 537

26 Literacy: Understanding Normal and Impaired Reading Development through Personalized Large-Scale Neurocomputational Models
JOHANNES C. ZIEGLER, CONRAD PERRY, AND MARCO ZORZI 554

27 Reasoning in Mathematical Development: Neurocognitive Foundations and Their Implications for the Classroom
JÉRÔME PRADO AND MARIE-LINE GARDES 566

28 Children's Scientific Reasoning Skills in Light of General Cognitive Development
PETER A. EDELSBRUNNER, RALPH SCHUMACHER, AND ELSBETH STERN 585

29 Working Memory Training: From the Laboratory to Schools
TORKEL KLINGBERG AND BRUNO SAUCE 606

30 Interventions for Improving Executive Functions during Development: Working Memory, Cognitive Flexibility, and Inhibition
NIKOLAUS STEINBEIS AND CLAIRE ROSALIE SMID 623

31 Curiosity-Driven Learning in Development: Computational Theories and Educational Applications
PIERRE-YVES OUDEYER 644

32 Neurocomputational Methods: From Models of Brain and Cognition to Artificial Intelligence in Education
MICHAEL S. C. THOMAS AND KASKA PORAYSKA-POMSTA 662

Index 688

The plate section can be found between pp. 534 and 535.

Figures

2.1 The neurone and the interneuronal connections through the synapse in the human brain *page* 30
2.2 Schematic representation of the hypothesis of epigenesis by selective stabilization of synapses 31
3.1 Structural changes of the brain during development 52
3.2 Cortical sulcation and cognitive efficiency 55
3.3 Structural changes of the white matter during development 58
3.4 Relationships between functional and structural markers of development 63
3.5 Top: Sulcus nomenclature and map of the standard sulcus interruptions. Bottom: The sulcus nomenclature projected on six different left hemispheres 69
3.6 White matter microstructure and connectivity with diffusion MRI 71
4.1 Neurovascular unit 86
4.2 Coronal slices of multimodal images of brain structure acquired in members of a birth cohort when they reached twenty years of age 89
4.3 The radial unit hypothesis 90
4.4 Estimated trajectories of cortical grey-matter and white-matter volumes during prenatal development, and cortical surface-area and thickness in post-natal development 91
4.5 Developmental trajectories of structural properties of white matter during infancy and early childhood, as revealed with multi-modal MRI 92
4.6 Myelin water fraction (MWF) and relaxation times in the developing brain 93
4.7 Total brain volume, as well as whole-brain (absolute) volumes of grey- and white-matter, derived from a combination of T1-weighted, T2-weighted and Proton Density-weighted images obtained in the NIH Paediatric Study 94
4.8 Decrease in cortical thickness between ten and twenty years of age 94
4.9 Cortical thickness and surface area in female (n = 509) and male (n = 479) adolescents 95
4.10 Virtual histology. Lateral (first row) and medial (second row) views of cortical thickness (Saguenay Youth Study) and gene expression levels (the Allen Human Brain Atlas) 96
4.11 Individual tractography results for the superior longitudinal fasciculus (orange), inferior longitudinal fasciculus (magenta), and corticospinal tracts (green) are shown in three representative healthy individuals at different ages 98
4.12 Volume of white matter (cerebral lobes) in female (n = 509) and male (n = 476) adolescents 99
4.13 Multimodal imaging of the human corpus callosum: a comparison with histology 100

II.1	Three-systems theory of the cognitive brain	145
10.1	Example items from the number sets task	233
11.1	Familiarization and test trials shown in the false-belief condition of Scott and Baillargeon. The order of the two test events was counterbalanced across infants	249
11.2	Familiarization and test trials shown in the deception condition of Scott et al.	250
12.1	Example of a moral dilemma used by Haidt and colleagues in their "moral dumbfounding" experiments	287
14.1	A simple neural network with three layers. Units are represented as circles, connection weights as arrows	319
14.2	The first five phases of a CC network with four input units and two output units. Frozen input weights are represented by solid lines, trainable weights by dashed lines	321
14.3	An SDCC network with two inputs, two layers of hidden units, and one output unit	322
14.4	Example of a KBCC network that first recruited a previously-learnt source network and then a single hidden unit	322
14.5	Example balance-scale problem, with four weights placed one peg from the fulcrum on the left side and three weights placed two pegs from the fulcrum on the right side	323
14.6	Proportion of LCA rule frequencies across stages, revealing overlapping waves in development	326
14.7	An example joint-attention event in word learning	327
14.8	Architecture and information flow in the TRACX2 neural-network model	329
16.1	Development of understanding of different types of symbolic numbers, including the age range in which development is most dramatic	364
16.2	The overlapping waves model	365
19.1	Performance on the NIH Toolbox DCCS test and the Toolbox Flanker Inhibitory Control ("Flanker EF") and Attention ("Flanker Attention") test across age groups	433
19.2	Neurocognitive processes (and structures) involved in deliberate fluid reasoning, intentional action, emotion regulation, and social function	435
23.1	(a) Sagittal (left), coronal (middle), and axial (right) views of frontostriatal white matter bundles of one random participant. (b) Sagittal (left), coronal (middle), and axial (right) views of areas of activation involved in cognitive control and risky decision-making	502
24.1	The triadic model	517
24.2	Overall strategy for testing the triadic neural systems model	523
24.3	Strategy for addressing Question 1, Step-1: Behavioral data reduction and organization	524
24.4	Strategy for addressing Question 1, Step-2 & Step-3: Supervised ML step	525
24.5	Two examples of solutions using classification tree algorithms	526

24.6 Strategy for Addressing Question 2, Step-1 to Step-3: Extraction of neural predictors, unsupervised ML clustering, and supervised ML to predict behavioral factors aF and vF 528
24.7 Fictitious results to Question 3: Decision trees across development 531
25.1 The intraparietal sulcus 542
25.2 Two scenarios where ANS acuity will be related to the highlighted topic if prerequisite topics are not entered as a covariate to topic performance 547
26.1 An implementation of the phonological decoding self-teaching hypothesis 557
26.2 Simulations of learning to read through phonological decoding and self-teaching 558
26.3 The use of decoding versus direct instruction as a function of reading skill 559
26.4 The individual-deficit simulation approach 560
26.5 Predicted versus actual reading performance 561
26.6 Predicting intervention outcomes through 'personalized' simulations 562
27.1 Example of materials used in the problem about transitive relations 578
27.2 Example of materials used in the patterning problem 578
27.3 Examples of line configurations with different values of n in the geometry problem 579
28.1 Setup of the pendulum task with the variables string length (two pendulums on the left: shorter; two pendulums on the right: longer) and heaviness of the weight (grey weights: lighter; black weights: heavier) 586
28.2 Can you help Paul to solve this riddle by providing your domain-general scientific reasoning skills? 588
28.3 Item demanding domain-specific scientific thinking 589
28.4 Main components of the SDDS model of scientific reasoning 590
28.5 Schematic depiction of significant associations (all with positive regression weights) found in multivariate models of Osterhaus et al. (a) and van der Graaf et al. (b) 599
31.1 Open-source baby robots 646
31.2 The playground experiment 649
31.3 Educational game used in Clement et al.: a scenario where elementary school children have to learn to manipulate money is used to teach them the decomposition of integer and decimal numbers 656
31.4 Example of the evolution of the zone-of-proximal development (ZPD) based on the empirical results of the student 657
32.1 Architecture of the Harm and Seidenberg model of reading, showing the specified representations (semantics, phonology, and orthography), pathways and directions of activation flow between them 670
32.2 Summary of dynamic causal modelling of functional magnetic resonance imaging data, showing which brain regions involved in reading appeared to causally drive other regions 671
32.3 Long and Aleven's skills meter bars indicating the level of student skill mastery 677

32.4 TARDIS scrutable OLM showing synchronized recordings of the learners interacting with the AI agents along with the interpretation of the learner's low-level social signals such as gaze patterns, gestures and voice activation in terms of higher-level judgements about the quality of those behaviours, e.g., energy in voice 678
32.5 Four conditions examined by Krach et al. 680
32.6 Regions associated with 'theory of mind' grow more active as the appearance of a technological opponent becomes more human-like, even when it is clearly not human 680
32.7 A child playing with the ECHOES agent through the multi-touch screen interface (Left). The agent points to a flower that it wants a child to pick and put in the basket in a bid for attention and interaction with the child (Right) 681

Tables

1.1 Stages of life *page* 16
5.1 Correlations between temperament measures at seven months and ANT scores at seven years 111
5.2 Brain attention networks, anatomy, dominant modulators, and genetic alleles 112
10.1 Sequence of tasks and approximate mean ages and ranges over four years 226
18.1 Overview of replication success of early belief understanding studies from Kulke and Rakoczy 415
23.1 Overview of paradigms to measure cognitive control and risky decision-making across adolescence 501
32.1 Components of two different types of computational model, those used to simulate cognitive mechanisms and those used as theoretically informed artificial intelligence tools to support learning and teaching 667

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Introduction

Cognitive Development Studies: From the History of Psychology to the Current Trends in Cognitive Sciences

Olivier Houdé and Grégoire Borst

The history of cognitive development studies began with the Greek philosopher Plato (Houdé, 2019)[1]. In the centre of Raphael's famous fresco, *The School of Athens* (1512) in the Vatican Museum in Rome, Italy, we see Plato (428–347 BC) and Aristotle (384–322 BC). Plato is pointing upwards to the 'Heaven of Ideas', while Aristotle, his pupil at the Academy, stretches his hand forward, symbolizing the earthly world. Indeed, for Aristotle (who does not believe in the Ideas as such), the general and the particular are transmitted here.

I.1 Step 1: The Ancient Roots of Innatism and/or Empiricism

For Plato, souls, thought of as 'immortal', have already contemplated the world of Ideas – i.e., the Good, the True, and the Beautiful – during their prenatal period. Birth disturbs this process. That is why 'the body is a tomb', according to the philosopher's expression. Nevertheless, thanks to a psychological phenomenon of *reminiscence*, triggered by the perception of concrete things in the sensible world (relations, numbers, and qualities), we can rediscover the innate Ideas. For example, Plato states that we grasp the idea of perfect equality with pieces of wood that are almost equal, but that the 'equal' itself does not reside in the pieces of wood. It is we ourselves who, from sensible objects, infer their essence. In the same way, when we look at six knucklebones, we cannot say that the number six is in any one of them, or in all of them together, since it is we ourselves who have made the connection between the Idea and the objects. Even Meno's slave, who appears in the dialogue of the same name, is able to deduce, starting from a right-angled triangle, Pythagoras's theorem: *the square of the hypotenuse is equal to the sum of the squares of the other two sides.* A century before Plato, the geometer Pythagoras (580–495 BC) had indeed demonstrated this theorem – a 'discovery of the True'.

According to Plato, these immutable Ideas are latent knowledge: they are within us from birth, without our knowing it. In this already subtle psychology, it is not a question of *ignorance* but of *latency*: dormant truths for which man seeks. It is at the cost of a mental effort, often requiring a whole education, that the innate Ideas can (re)appear. They are recalled and reactivated. Hence the importance, from a Platonic perspective, of the *maieutics* of Socrates, the art of 'giving birth to minds' (Socrates' mother was apparently a midwife), causing doubt and astonishment in one's interlocutor. Plato thus underlines the educational role of the social environment. However, learning is not a matter of filling the mind, seen as an empty *tabula rasa* ('blank slate'),

[1] Even before Socrate and Plato, there were pre-Socratic thinkers whose conceptions already looked forward to cognitive development, such as Parmenides (sixth–fifth century BC), who thought that perception was an illusion, and indeed a fraud, compared to thought.

by means of mere sensation, as it is in Aristotle, and as the young Theaetetus, an empiricist, suggests in a Socratic dialogue narrated by Plato. On the contrary, the Ideas are already present from birth, as a pre-existing cognitive store, a capital sum of reason that can be reactivated. Here we identify the ancient root of the *rationalist and innatist* trend that stretched for two millennia, right up to the 'core knowledge' of the infant cognitive psychology, according to Elizabeth S. Spelke (2000), via Descartes, Kant, Noam Chomsky (1975) and Jerry Fodor (1983).

Plato also anticipated the cerebrocentric approach: he already located the rational part of the soul (mind, intellect, reason) in the brain, whereas his pupil Aristotle, at the School of Athens, still located it in the heart (cardiocentrism). Plato, however, placed desire and its impulses in the lower abdomen, and the will in the heart. We now know that everything is in the brain (Changeux, 1985, 2012), including the mapping of desires and emotions through feelings, even though the brain's relations with the whole body are always intimate and continuous (Damasio, 1994, 2018).

Let us return to Raphael's fresco, *The School of Athens*, and Aristotle's place in it. According to Aristotle, at birth we are a *tabula rasa*. Thus, it is only within things themselves that their essence is to be found, and not in a superior Idea that transcends them.

What interests Aristotle is the discovery of *what the earthly world is made of*, the 'genera of being': quality, quantity, relation, place, time, action, etc., and their subdivisions. That is why his approach is encyclopaedic and known as 'systematic': he classifies diversity. Aristotle has thus influenced separate branches of knowledge: psychology, logic and zoology, all of which include the *logos*, implying reason and language. Aristotle distinguishes between external objects, mental images and their communication by words; but in order to understand the world and to speak correctly of it, he wishes to establish a rigorous logical reasoning that connects objects to words: a science of sciences. Therefore, he is important to philosophy, certainly, but he also prefigures the psychology of reasoning. Indeed, we owe him the discovery of the 'syllogism'.

Aristotle used logic to detect and refute the errors of reasoning, the sophisms and paralogisms, of his contemporaries in Greece (sophisms corresponds to the fake news of today, Lazer et al., 2018). Here lies the ancient root of the study of the cognitive biases of reasoning (Evans, 1989; Kahneman, 2011) and the psychological forms of syllogisms – the rules of mental logic (Braine & O'Brien, 1998) and visual-spatial models (Johnson-Laird, 2001).

A fundamental problem nevertheless remained for Aristotle: if the science of syllogisms aims at the objective knowledge of the world, how do we obtain the first true propositions, which do not themselves result from a prior deduction? Indeed, we can produce an argument whose conclusion (c) is logical (valid) but has a premise (either a or b) that is false. For example, if (a) *all men are immortal* and (b) *Socrates is a man*, then (c) *Socrates is immortal*. However, Socrates, a Greek philosopher, and the master of Plato, was mortal! How can we ensure that the initial proposals are trustworthy? This is where induction and empiricism find their place in Aristotle's theory. Not believing in the Ideas of Plato (as starting principles that are inside us without our knowledge), he appeals to a *sure and certain* faculty of recognition and judgment, which results from perception: there is no Idea without such a prior impression. Psychological empiricism is therefore inevitable. The *tabula rasa* is a corollary of the science of syllogisms.

According to Aristotle, the senses and induction lie at the origin of everything. From sensation arises memory, from memory

comes experience and from *reasoned* experience (syllogism) comes the conception of the universal. It is through this inductive empirical process alone – without needing first to attain the pre-existing Ideas – that we can grasp the indemonstrable elements or axioms that are seen as trustworthy, appropriate and self-evident. From these the necessary deductions will then follow.

At birth, the intellect is similar to a tablet on which nothing is currently written, argues Aristotle in his *Peri psychès* (*On the Soul*) – arguably the first comprehensive treatise on psychology in history (just as the *Organon* was for logic). He describes in fine detail how a sensible object causes a specialized activity of the cognitive function. The latter does not receive the sensible form 'from without', but re-creates it 'from within', from its own powers, when it is affected from without. This nuanced empiricism (internal re-creation) already foreshadows the cognitive process of assimilation-accommodation described by Jean Piaget (2015; Piaget & Inhelder, 1969). According to Plato's pupil, the powers of the cognitive function come from a general breath of life, 'the soul', shared by all animate beings: plants, endowed with faculties of nourishment and growth; animals, endowed with faculties of motion and perception in addition to the preceding ones; and humans, who also benefit from the cognitive faculties of thought, of reasoning. In this way, three souls intertwined by epigenesis form a hierarchy in man: vegetative, sensitive and intellective. Human intelligence is an extension of biological adaptation, once again prefiguring the work of Piaget. The living world of Aristotle, however, is fixed: it is a *scala naturae* ('scale of beings') created by God, in which man dominates an eternal universe (like that of Socrates and Plato), not a transformist world in evolution of the kind later discovered by Buffon, Lamarck and Darwin. Unable to imagine this revolution in biology, Aristotle, as a good logician, wished to avoid an infinite regress: so, he formulated the hypothesis of a god being, the cause of all, and himself uncaused – in other words, the 'first cause'.

I.2 Step 2: The Renaissance of Innatism and/or Empiricism

According to the French philosopher René Descartes (1596–1650), men are composed of two natures: the body and the soul, the latter being specifically defined as a 'thinking substance'. He compares the body to a pipe organ, in which animal spirits act like the air between the ducts in some pipes. Body and soul may well be joined together and united by the pineal gland in the brain (the epiphysis), but only the body is a machine. This difference in nature (or substance) between the soul and the body constitutes what is called 'Cartesian dualism', which left a permanent mark on psychology: the body was the domain of physiologists and doctors, the soul was the domain of psychologists.

Descartes was an innatist, like Plato. Indeed, his thought drew on the *rationalism and innatism* of Ancient times. The Cartesian method and rules were designed to discover the Ideas and properly develop them. In Descartes, human doubt stems from an imperfection in method, whereas God is perfection. Thus, to the question 'From where do we get this precious treasure that is our intelligence?', Descartes, in his *Treatise on Man* (1664), answers with the self-evident and apparently inescapable truth: God has deposited in our minds, from birth, clear and distinct logical and mathematical ideas, the core of human intelligence. An infant is thus 'potentially intelligent' (a concept very common these days) but is intelligent thanks to God's gift. This divine explanation would be shattered by nineteenth-century biology and Darwin's description of a

natural biological evolution of animal and human intelligence – excluding God from the explanation.

In the same way, the Cartesian dualism between the mind (or soul) and the body would be shattered by the neurosciences of the twentieth century – prefigured in Ancient times by the Greek physicians Herophilus and Galen – through the mechanistic study of the brain as an organ of thought, especially using brain imaging techniques. Moreover, Descartes' correlative theory of machine-animals was directly challenged by La Mettrie's *Man a Machine* (1748), which, though written in the eighteenth century, already foreshadowed contemporary artificial intelligence (the brain-machine), a branch of computer science. Finally, even though Descartes wrote a *Treatise on the Passions*, it was the dualism between the soul and the body that Damasio would denounce in *Descartes' Error* (1994), demonstrating, with the help of contemporary neuroscience and the 'somatic marker theory' that we think with our bodies and our emotions in a system of generalized equilibrium called homeostasis (Damasio & Carvalho, 2013). Damasio (2003) agreed more with Baruch Spinoza (1632–1677), who, somewhat later in the seventeenth century, came closer to modern neurobiology than had Descartes by bringing mind and body together, ascribing to the emotions a central role in human survival and culture.

After Descartes, at the Age of Enlightenment, a new and famous cognitive problem added new fuel to the debate between the advocates of innatism and empiricism: the 'Molyneux's problem', named after an Irish scholar who asked the English philosopher and physician John Locke (1632–1704) whether a man who was born blind, then grew old and was cured, would be able to distinguish by sight, without touching them, a sphere and a cube placed on a table in front of him – assuming that he had previously learned to distinguish these two objects by touch alone. Molyneux (1688 [1978]) himself believed that he would not. Locke was of the same opinion, for, according to him, visual ideas and tactile ideas were acquired independently by sense experience; and in this case, *the necessary association between the two* had not been made. It is now known that intermodal touch-vision transfer is possible at a central level in the brain, even in an infant, so the answer to Molyneux's problem is therefore 'yes' from the point of view of the contemporary psychology of cognitive development. But the problem fascinated the eighteenth century, which debated opposite points of view.

Molyneux's problem, apparently technical and limited in scope, is historically important because it corresponded to a more general key question, which aroused lively debate in the eighteenth century: what is the role of the environment and experience in the construction of our knowledge? Locke, who inaugurated the Enlightenment in England, answered this question with his empirical philosophy: knowledge results directly from the experience of the reality of the senses. Our ideas are not of divine origin (as in Descartes) but come from perception. It is out in the world that they are found; they do not lie innate within us. Otherwise, how can we explain the diversity of men, the ignorance of children, savages, idiots, and so on? Locke was a paediatrician and tutor to the children of an important politician in England, Lord Ashley; what Locke was aiming for was the *upbringing of a young English gentleman* but he also understood how each culture and each epoch has its own infancy. Hence his psychology of the child and of education, a corollary of his empiricism. Locke, like Montaigne, was struck by the weight of custom and habit, and the role of circumstances. From his observations, from his travels, from the stories he heard, he

deduced that human understanding was a matter of environment, of education, and a 'reflection of nature'.

Thus, in his *Essay Concerning Human Understanding* (1690), Locke refuted Descartes' innatist psychology (and all those that had preceded it ever since Plato). Inspired rather by Aristotle, he made the human mind a *tabula rasa* on which sensations are formed during childhood, shaping ideas that associate and combine, from the simple to the complex. He defined the idea as 'whatsoever is the Object of the Understanding when a Man thinks' (a representation). According to Locke, experience can be applied either to the external objects of the environment (association between *sensation* and idea) or to the internal operations of the mind (reflection through the association of ideas). He thought that ideas couldn't be separated from consciousness. Becoming aware of ideas is an awakening of consciousness. Education to ideas is an awakening of consciousness.

In the problem of Molyneux, the man born blind possesses the conscious experience of an association between the tactile sensation and the idea of a sphere, but not between the visual sensation and the same sphere. So, nothing in his previous experience allows him to associate the two in his mind (hence Locke's negative answer). The missing mechanism, very simply, was *association*. What Locke wanted to discover, in line with the Newtonian scientific spirit and in opposition to Descartes, was a simple psychological mechanism, a functioning law that regulated the mind, the mental realm. This was the association between sensations and ideas, and the association between ideas themselves. Thus, association governed the 'world of ideas', that is, the world of psychology, like the mechanism of gravitation theorized by Newton, whose physics governed the fall of bodies and the relations between celestial bodies. Locke's work acceded to a request Newton had expressed in the *Principia* (1687): finding for the mind, as he had done for space, one, and only one, universal principle of operation.

However, this principle or mechanism of association of ideas necessarily implies a psychology of education. This is what Locke proposes in *Some Thoughts Concerning Education* (1693), in which he outlines a psychology of the child. With a clean slate as his starting point, Locke understands that the powers of the mind will demand social incentives and models if they are to develop. The *mechanism* needs to be educated, and he insists on the role of imitation and play. A properly understood education must use games that are both free and challenging, but it must reserve a place for spontaneous imitation. What fascinates Locke in children is their drive and enthusiasm, that childlike zest that bursts forth in action, play and even schoolwork. In action, as in play, this was, in his view, all an expression of *freedom*.

Locke had all it needed to be a child psychologist, but he did no more than touch on the essential problem: that of mental structures or frameworks. This was the main issue that Anglo-Saxon empiricism failed to address: beyond the mental content of knowledge (ideas), is there an active centre of the mind, a set of structures to which ideas cling? Locke came close to this when, in line with Aristotle, he described the child's powers of both feeling and reflecting. However, he said nothing about how these sentient and cognitive powers are structured (or structure themselves) in the mind. Returning to Descartes' innatism, Emmanuel Kant (1724–1804) will say that these powers were already in existence: for him, they were an *a priori* of understanding.

Following on from Locke, David Hume (1711–1776), a Scottish philosopher who was an acquaintance of Rousseau, completed the project of empiricism in the same scientific,

mechanistic and Newtonian spirit. He insisted on our faculty of *imagination* that worked on the basis of sensations: we *imagine* ideas, which involves a certain margin of uncertainty. In order to understand how the very powerful human imagination associates ideas with each other on the basis of experience by bonding them together (by 'attraction' as Newton would have put it), Hume described more precisely than Locke three sub-processes: (a) the spatial or temporal *contiguity* of objects in reality, a contiguity which configures our memory and the mental evocation of ideas; (b) the *resemblance* of each copy to the general idea; (c) the *cause and effect relationship* that underpins our belief system and our practical knowledge.

According to Hume, these processes of 'assembly' can operate incredibly fast in the human mind. His empiricism was original, very cognitive in nature (memory, beliefs, etc.) and even prefigured Daniel Kahneman's *system 1* today (Kahneman, 2011).

In France, Étienne Bonnot de Condillac (1714–1780) formulated a so-called sensualist version of Anglo-Saxon empiricism in his *Essay on the Origin of Human Knowledge* (1746): all our understanding, from perception to judgment, is *and must be* derived wholly from sensations. This means that we need to reform and purify our language so that it will organize sensations in such a way as to impose a precise correlation between words and things. That is what Condillac, a member of the French Academy, advocated to the scholars and philosophers of his time: they needed to express their knowledge in a correct, pure, clear language so that everyone would be able to grasp that knowledge. This was also the aim of Diderot in the *Encyclopédie*. Condillac wrote his *Grammar*, *The Art of Writing*, *The Art of Thinking*, *The Art of Reasoning*, and so on with this aim. In his main work, *The Treatise on Sensations* (1754), he explained that the understanding was like an inert 'statue' that resided within us before existing in the outside world; contact with the world through the senses gave this statue life in the following order: smell, hearing, taste, sight, touch. The ideas derived from these five senses were simply sensations designated by words that represented things. These sensations developed and combined, thereby shaping the understanding. However, this still leaves the problem of the mysterious inert statue from which it all began!

From Locke to Condillac, empiricism assumed various different forms in order to promote the same general psychology that marked a revolt against the Cartesian *cogito*: for the empiricists, the environment shaped reason. Nevertheless, the question of the powers of the understanding touched on by Locke in his psychology of the child, and the 'inert statue' within us in Condillac, reveals the difficulty empiricism encounters in giving an account of the frameworks of the mind, which can structure and perhaps even precede the experience of sensations. Without this, the associations of ideas by the empirical connections of contiguity, resemblance, or cause and effect, as in Hume, remain contingent, 'scattered around' by circumstances without any real active centre (self, or *cogito*, in Descartes' phrase). This is where Leibniz, Wolff and Kant contributed to the debate.

The German mathematician Gottfried Leibniz (1646–1716) refuted Locke's thesis on the non-innate nature of ideas point by point. His critique promoted the view that there are Ideas that exist *independently of us*. According to Leibniz, the understanding is innate and allows us to process the data from experience through 'necessary ideas': that is why he gives the answer 'yes' to Molyneux's problem. By definition, these ideas are not contingent; that is, they do not depend on environmental circumstances, contrary to what the empiricists

believed. Moreover, Leibniz observes that when the understanding, endowed with its necessary ideas (the self, the *cogito*), meets the perceptual world, representations emerge, which are not always very clear, mixed with an infinity of 'little perceptions' that can confuse consciousness and reflection and sometimes even elude them altogether. This view was still quite different from that of Locke, who supposed that ideas derived from the senses could not be separated from consciousness. In the end, one has the impression that, for Leibniz, the role of the environment was not positive or, at least, not as able to structure mental life as the empiricists seemed to suggest.

A pupil of Leibniz, Christian Wolff (1678–1754), then published a two-part treatise on psychology *Psychologia empirica* (1732) and *Psychologia rationalis* (1734) that forged a remarkable methodological synthesis, paving the way for Kant. As these titles indicate, Wolff distinguishes between two types of psychology. One is *empirical*: it needs to be based on external or internal observation, through introspection, and to lay bare the laws of human conduct and the faculties of the soul by measurement and calculation, like in physics. Wolff called this very prophetically 'psychometry', heralding Fechner's psychophysics in the next century. The other psychology is *rational* and belongs to pure reason: it enables us to determine *a priori*, by reasoning, as in algebra or geometry, what the faculties of the soul must be. Wolff advocates combining these two psychologies; the second kind is also able to feed on data known *a posteriori*. Thus, almost all-future psychology found itself already defined.

After Leibniz and Wolff, their disciple Kant formulated propositions that are still current in today's psychology, in line with a tradition that went back to Plato via Descartes. He said that pure concepts exist in us innately, as mental frameworks or 'categories of the understanding'; these do not come from the sensible world (the environment of the empiricists). Kant, following Wolff, gives the name 'pure reason' to so-called transcendental, metaphysical knowledge, superior to and outside of the world. This exists *a priori*, independently of our sensations. These *necessary and universal* cognitive principles relating to space, time, number, etc., are in us from birth, but only sensible experience, from infancy to adulthood, can *reveal them*. Thus, according to Kant, reason neither alone (rationalism) nor sensations alone (empiricism) make it possible to know the world. The intermediary here is the 'schema', which links pure, innate concepts with *intuitions*, for example intuitions of space, time, number, etc. that are linked to sensible experience and its representations. Thus, the schemas, or frameworks, of the mind allow us to make judgments about reality. This notion of a schema (or scheme) would be made famous in the twentieth century by Piaget, who made it the basic unit in his constructivist (non-innatist) theory of intelligence in the child. Other current programmes in the non-Piagetian cognitive sciences are more innatist and follow Kant in considering the possibility of *a priori* mental frameworks which sensible experience reveals from infancy to adulthood. This is the case with Dehaene and Brannon's (2010) view: 'Space, Time, and Number: A Kantian Research Program'.

The path travelled since the Renaissance shows that the history of psychology has been punctuated by a cognitive reform. In the seventeenth century, cognitive reform began with Descartes, the discoverer of the *cogito*, and Pascal, the critic of reason, who promoted *finesse* against *geometry*; but he was still reckoning without a mechanistic science (that of Copernicus, Galileo and Newton) that had not yet affected psychology. We had to wait for empiricists like Locke and Hume *to move on from the laws of space to those of the mind*. Like

nature, the mind itself could now be something to be observed, analysed and understood from an external scientific perspective. The new explanation, however, still lacked 'frameworks', and Kant restored them: pure concepts and innatism, *schemas for sensible experience*. Kant thus saw rationalism and empiricism as two sides of the same coin. While his approach was metaphysical, the repercussions of the *a priori* on psychology were still great, forming a kind of tradition leading from Plato to Descartes and then on to the innatism of the current cognitive sciences.

I.3 Step 3: From Innatism/Empiricism Duality to Jean Piaget's Constructivism

In the nineteenth century, the English naturalist Charles Darwin (1809–1882), in *The Origin of Species* (1859), formulated the hypothesis – now accepted – of a general mechanism of *natural selection* that applies to *populations* and is based on a two-fold principle. The first principle is the *variation* of characteristics, generation after generation (the genetic origin of this random variation would be understood only after Darwin, thanks to Mendel's laws). The second principle is the *selection* by survival and reproduction of those who have (by chance) the combination of characteristics best suited to their environment. The effect of the environment is therefore indirect, and there is no longer, as in Lamarck, a heredity of acquired characteristics (by daily habit). To this, Darwin adds a world-shattering new idea: *all* living things have a common ancestry, because life on earth has a unique origin, from bacteria and blue algae to man!

Thus Darwin introduced into science and psychology the idea of a *natural* evolution of animal and human intelligence or cognition through *phylogenesis* or the evolution of species, a process lasting millions of years, in which matter, life and thought are intertwined – thus excluding God from scientific explanation (and, by the same token, ruling out the divine-innate ideas of Descartes and Plato). In *The Expression of Emotions in Man and Animals* (1872), Darwin described in detail studies of the expressions on the face and the emergence of language in children. The baby he observed was his own son, Doddy Darwin. This was the century of the first monographs devoted to children: the French historian Hippolyte Taine (1828–1893) and the English physiologist William Preyer (1841–1897) both reported observations taken from their own children. The Americans Stanley Hall and Arnold Gesell would later develop methods that are more systematic.

In the twentieth century, this idea of an evolution of intelligence was taken up again in the study of *ontogeny* – the idea that, from infancy to adulthood, body and mind evolve. The Swiss Jean Piaget (1896–1980) developed this in the psychology of the child's cognitive development and the French Changeux in neurobiology (his 'neural-mental Darwinism'). According to the latter theory, Darwin's variation–selection mechanisms also operate in the brain itself, affecting cognitive representations within *neuronal populations* – a theory that Changeux (1983, 2012) shares with the American Nobel Laureate in Physiology or Medicine Gerald Edelman (1929–2014). In fact, all scientific psychology has to consider two timescales: *phylogenesis*, or the evolution of species (Darwin), and *ontogenesis*, that is, development from infant to adult (including embryogenesis). In addition, there is a cognitive 'microgenesis', which corresponds to the much shorter time taken up by learning or by the brain when solving a task: months, days, hours, minutes, fractions of seconds, i.e., milliseconds.

In the twentieth century, the most dominant figure in the study of cognitive development

was Piaget. His very new approach was the basis of genetic (in the sense of *ontogenesis*, not genomic) psychology and epistemology. Piaget's contribution to the history of psychology and ideas in general was to produce a real synthesis between empiricism (from Aristotle to Locke and Hume) and innatism (from Plato to Descartes and Kant). He rejected both trends as too simplistic and proposed a third intermediate way: *constructivism*. In his experimental and clinical studies involving children, he demonstrated the construction of psychological structures, from the stage of sensorimotor *schemes* (as in Kant) in the infant to the stages of concrete operations in the child and then the formal (abstract) operations in the adolescent. This comprised the genesis of logico-mathematical intelligence, from intuitive and illogical levels in infants and preschool children to the logical levels in school children and adolescents, a view now revised by post-Piaget thinkers. Others supplemented it by focusing on the more emotional, social and cultural origins of human cognition: namely the Russian psychologist Lev Vygotski (1896–1934), opening the way to the American psychologists Jerome Bruner (1915–2016) and Michael Tomasello (b. 1950).

Challenging both the empiricism of Locke, Hume and Condillac, who believed everything derives from experience through association and practice, and innatism (the opposing theory), which explains everything through innate structures (see Plato and Descartes, but also Kant and his *a priori* forms of sensitivity), Piaget proposed an intermediate theory, called 'constructivism'. It holds that intellectual structures, i.e. our thoughts, our mental operations, have a genesis specific to them (cognitive ontogenesis). From birth to adulthood, they are gradually *constructed*, stage by stage (like going upstairs one-step at a time), in the context of the interaction between an individual and his or her environment – or, in more biological terms, between a body and its environment. In this interaction, what is essential for Piaget, is the *action* of a child on objects that surround it (exploring, handling and 'experimentation'), a concept very different from the idea of 'passive' learning (association and practice) specific to empiricism.

I.4 Step 4: Neo- and Neuro-constructivism in Cognitive Sciences

The purpose of this *Cambridge Handbook of Cognitive Development* is to provide comprehensive and detailed work on cognitive and brain development, since the seminal work of Piaget. That is for 40 years (1980–2020).

During the last decades, detailed behavioural studies have shown that Piaget underestimated the cognitive and logical (even scientific) capabilities of infants, as shown by the American psychologists Renée Baillargeon, Alison Gopnik, Elisabeth Spelke and Karen Wynn for example (Gopnik, 1999, 2012). Conversely, Piaget overestimated the cognitive capabilities of adolescents and adults, which are often biased by illogical intuitions and overlearned strategies (or heuristics) they fail to inhibit, as shown by the Israeli-American psychologist Daniel Kahneman (2011), Nobel laureate in Economic sciences in 2002 – and other authors, namely the British Jonathan Evans (1989, 2003).

Therefore, cognitive developmental psychologists are now facing a big paradox: the early smart skills of infants and the late illogical ones of older children and even adults. The new challenge is therefore to account not only for an incremental stage-by-stage process like in Piaget's approach, but also for a non-linear dynamical system of growth (Siegler, 1996). Within such a system, competing strategies, i.e., intuitive ones (heuristics) and logical ones (exact algorithms), may occur in the

developing brain with different weights at any point in time, depending on the context and individual differences. This dynamical modelling introduces less regular developmental curves containing perturbations, bursts and collapses. Often, in some contexts, children fail to inhibit misleading heuristics (Borst & Houdé, 2014; Houdé, 2019), which explains these developmental irregularities, formerly called horizontal and vertical 'décalages' by Piaget ('only to save face', because it was a very strong objection to his incremental and structuralist theory). In fact, 'décalages' are not the exception, as Piaget thought, but the rule!

The current non-linear and dynamical approach better matches with the complexity of cognitive development that all the parents and teachers watch every day. It also points to the brain attentional and cognitive-control mechanisms as the key factors of development.

The general aim of the handbook is to present, illustrate and summarise this new epistemology of cognitive and brain development that we call 'neo- and neuro-constructivism' (in line with Mareschal & Johnson, 2007). 'Neuro' because of, after Piaget's time, the development of brain imaging techniques and their use in experimental psychology conducted to study both brain construction and cognitive development. Within this co-construction, we must also now consider the multiple mechanisms by which genes may influence behaviour, having in mind, as Changeux states in this book (Chapter 2), that:

> the human brain is, neither John Locke's blank slate (*tabula rasa*) deprived of any pre-existing innate structure – or, in a modern language, a random network of undifferentiated neurons fully instructed by experience – nor a fully genetically determined, irrevocably hard-wired neuronal architecture … It is a unique compromise between an eminently variable, intrinsically rich, connectivity and a set of species-specific, genetically determinated, rules, which unambiguously make our brain that of *Homo sapiens*.

To achieve this general 'neo- and neuro-constructivism' aim, we have invited current leading scholars from neurobiology, developmental cognitive neuroscience, experimental child psychology and computational modelling.

I.5 Overview of the Handbook Contents

An opening section (Part I) covers the neurobiological constraints and laws of cognitive development: *how the brain constructs cognition.* Damasio and Damasio (Chapter 1) describe and explain how life regulation and feelings motivate the cultural mind, from the dawn of life to humans today, through *phylogenesis*. Then, in Chapter 2, Changeux explores the key principles of *epigenesis*, synapse selection, cultural imprints and human brain development, from molecules to cognition. Chapter 3, by Cachia, Mangin and Dubois, presents the current mapping of the human brain from the preterm period to infancy using 3D magnetic resonance imaging (MRI). Next, Chapter 4, by Paus, continues this description by the human brain development and maturation processes from infancy to adolescence. Then, Posner and Rothbart (Chapter 5) add an important focus on genetic and experiential factors in brain development using the examples of executive attention and self-regulation. Finally, Luna et al. ends this Part I developmental description by the brain basis, underlying the transition from adolescence to adulthood in Chapter 6.

The following section (Part II) covers the fundamentals of cognitive development, from infancy to adolescence and young adulthood. A first subsection covers the roots of thinking in infancy. It starts with Vauclair's description

of the differences between humans (from infancy), great apes and monkeys in cognition (Chapter 7). Then, the following chapters explore infant physical reasoning about objects (Lin, Stavans, and Baillargeon, Chapter 8), categorization (Rakison, Benton, and Dinh, Chapter 9), numerical cognition (vanMarle, Chapter 10), theory of mind (Scott, Roby, and Baillargeon, Chapter 11), social-moral cognition (Falck, Strickland, and Jacob, Chapter 12), and scientific thinking and reasoning (Goddu and Gopnik, Chapter 13). A final chapter deals with computational approaches to cognitive development in infancy (Nobandegani and Shultz, Chapter 14).

A second subsection covers cognitive development during childhood and adolescence. It starts with language and categorization in young children (Gelman, Chapter 15). Then, the authors explore the question of numerical strategy choices in preschool and schoolchildren (Siegler, Chapter 16) and the role of inhibitory control in these choices (Roell, Houdé, Borst and Viarouge, Chapter 17). The following chapters present several other facets of cognitive development during childhood and adolescence. In Chapter 18, Perner, Kulakova and Rafetseder examine theory of mind and counterfactual reasoning together. In Chapters 19 and 20, Happaney and Zelazo, and Chevalier and Blaye describe the general role of executive functions and flexibility. In Chapter 21, Frey and De Neys explore the reasoning biases through dual processes (intuition versus logic). Then Chalik, Misch and Dunhan deal with social cognitive development in Chapter 22. Finally, Chapters 23 and 24 examine the functioning of cognitive control, decision-making and motivation in children and adolescents (Blankenstein, Peper and Crone; Ernst, Gowin and Gaillard).

The last section (Part III) covers the education and school-learning domains. Our aim here is to formulate a set of cognitive principles for evidence-based education (Lau and Ansari, Chapter 25). Ziegler, Perry and Zorzi start with the question of literacy and normal versus impaired reading, in Chapter 26. Chapter 27 explores numerical cognition and mathematical reasoning in the classroom (Prado and Gardes). Then, Edelsbrunner, Schumacher and Stern examine scientific reasoning skills in light of general cognitive development, in Chapter 28. Chapter 29, by Klingberg and Sauce, presents a focus on working memory training, from the laboratory to school. In addition, Steinbeis and de Smid draw a more general overview of working memory, cognitive flexibility and inhibition trainings or interventions at school, in Chapter 30. Chapter 31, by Oudeyer, deals with curiosity-driven learning and opens the field to developmental robotics. Finally, Thomas and Porayska-Pomsta show the power of neurocomputational methods in education, Chapter 32.

In each section, from Part I to Part III, beyond the specific cognitive or school domains, the authors analyse and discuss in depth the more general processes at the root of brain and cognitive development.

So, from Plato to neuroscientists today, beyond the outdated opposition between innatism and empiricism, we hold in hand for the twenty-first century a new and rich neuroconstructivist epistemology of cognitive development brilliantly illustrated by the contributions of the experts of this Handbook.

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Part I

Neurobiological Constraints and Laws of Cognitive Development

1 How Life Regulation and Feelings Motivate the Cultural Mind

A Neurobiological Account

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1.1 Sociocultural Phenomena Are Not Restricted to Humans

Few ideas are in greater need of correction than the common-sense notion that cultures are born exclusively of the human mind and constitute an entire novelty in the history of life. Somewhat distractedly and with the usual high regard that humans reserve for themselves, cultures are frequently thought of as purely human creations. For example, behaviors that result in moral regulation or in social and political organization, as well as activities tied to economical exchanges or the management of illness, are assumed to have been constructed by the grace of standard and explicit human intelligence. The human contemplation of facts, the process of reasoning over those facts, and the consequent knowledge writ large would have been the engines behind the production of cultures. Fine human cognition and sound human reasoning would explain the newness of such phenomena.

One understands why this scenario may be plausible, at first pass, and yet it is far removed from reality in numerous ways. Here are the facts. Simple organisms with only one cell and not even a nucleus, of the kind present at the dawn of life, around 4 billion years ago, already exhibited sociality. Groups of such organisms would band together to assist each other and improve their lot, but such groups could also fight other groups and compete with them for access to vital resources (Diggle et al., 2007; Hughes & Sperandio, 2008; Jousset et al., 2013; Naviaux, 2014; Nealson & Hastings, 2006; Persat et al., 2015; Rainey & Rainey, 2003). The social behaviors already included traits that resemble, in their essence, critical features of what, today, we designate as human cultures. It is an established fact that complex social arrangements began to appear in evolution long before multicellular organisms did, in fact, long before there were nervous systems, even simple ones (Table 1.1). As for cultural phenomena unquestionably worthy of that name, they certainly began to occur before there were vertebrates on the planet, as can be demonstrated by the elaborate compendium of social insect behaviors. The organization and the variety of roles played by bees and ants in their hives and nests is, for all practical purposes, "cultural" (Wilson, 2012).

Lest our statements be dismissed as crude and naïve, it is important to acknowledge that the degree of complexity of the non-human social behaviors to which we are referring is no match for the current social behavior of humans. But the critical point we are trying to establish here is that, nonetheless, the contours of the fundamental kinds of such non-human social behaviors resemble those of human cultures and that the fundamental consequences of such human and non-human behaviors are also comparable (Damasio, 2018). They revolve around the issue of maintaining the life process within the bounds set by homeostatic regulation. It is critical for our

Table 1.1. *Stages of life*

Beginning of Earth	around 4.5 billion years
Chemistry and Protocells	4.0 to 3.8 billion years
First Cells	3.8 to 3.7 billion years
Eukaryotic Cells	2 billion years
Multicellular Organisms	700 to 600 million years
Nervous Systems	around 500 million years

argument that the non-human social behaviors be part of a manifestly intelligent struggle to maintain conditions of existence compatible with the optimization of the life process and conducive to survival. In other words, the later appearance and subsequent development of human cultures in the history of life has had a main consequence: assisting and promoting the life-regulation processes whose ensemble is known as *homeostasis*. But that process of assisting and promoting life regulation was already established, practiced, and ongoing long before humans made their debut in evolution. The main consequence of non-human complex social behaviors – those behaviors we like to designate as "pro-cultural" – has long been the assisting and promoting of homeostasis. In a curious way, terms such as "pro-cultural," "pre-cultural," and even "cultural" simply mean "pro-homeostatic."

It is also important to acknowledge that non-human social behaviors developed and have continued to operate and evolve in non-human species, in the absence of conscious mental processes and in the absence of overt, explicit, deliberate creativity. In the vast majority of non-human organisms, the "author" responsible for the "pre-cultural behaviors" is the life process itself. That "leading responsible author" is not – as is obvious in the case of humans and of a number of animal species – the singular mind of one individual, conscious of a specific need and capable of reasoning creatively that the need can potentially be met by carrying out a specific action. Here is an illustration of our idea: when humans began to make use of warfare and also of pacts of cooperation, they were identifying problems and implementing solutions thanks to the fact that they were conscious and that their minds were capable of image-based reasoning. For all intents and purposes, they were guided by their creativity. But were humans really "inventing" solutions from scratch or were they, more plausibly, adjusting and perfecting pre-existing and quite competent solutions that their minds were now revealing and making explicit? After all, un-minded, non-conscious, and quite humble non-humans had responded comparably to such situations, long before, in the history of evolution. They had generated similar solutions in the absence of overt reasoning, in a manner that does not even qualify for "intuition," given how spontaneously and directly the actions appeared to spring fully formed from the living organism.

What was the essential mechanism, the "spring," as it were? It was, we believe, the *non-explicit*, *hidden sensing*, at any given moment, of the *degree of success or failure* of *internal life operations*, combined with the *concomitant and equally hidden sensing of the external conditions, at that moment*. We believe that using such terms as "mind" or "consciousness" for this process of non-explicit, non-imagetic sensing is not appropriate, although thinkers whose work we respect do so; that is the case with Arthur Reber (see Reber, 2019). We have begun to use the term "sentience" to describe such evaluative conditions, leaving out the vaguely conscious connotation associated with the word sentience. In brief, the "evaluation" of the life situation, no matter how covert, was responsible for adjusting life regulation, as needed, in keeping with the equally unstated, covert goal of maintaining the life of the organism. From a varied menu

of many possibilities of response and after several of them were tried out, effective responses were selected and adopted for the future.

What was the overall mechanism behind the discovery of effective solutions? "Why," "how," and by "whom" were solutions found and implemented?

In our current perspective, in most living creatures and for most of the duration of life on earth, the process of creating complex social behaviors has not been conscious, has not been minded or reasoned, and has not depended on the presence of nervous systems. Matters proceeded as outlined above based on "sentience" and on a fundamental "competence" that adjusted life parameters accordingly. We also believe, however, that once nervous systems appeared in the biological scene a novel and multi-pronged development began to loom. We propose that there were several fundamental steps in that development.

The first step consisted of the development of *feeling*, which, as we will discover in Section 1.3, amounts to *a mental experience of the state of life in the respective organism*. This is the de facto initiation of the grand process of consciousness. In order to do their job – advise their owner of the conditions of the organism – *feelings have to be conscious*, they have to pertain, unequivocally, to *their* specific organism. If feelings had not been *necessarily* conscious, their practical consequences would have been negligible. For example, a situation of pain would not have constituted a protective defense, nor would pleasure have been an incentive. We know today that, by and large, feelings are achieved by interactions of non-neural and neural parts of the organism. We address the physiology of feeling in Section 1.3.

The second step was the emergence of *consciousness proper*, concerning not only the organism but plenty of other objects, actions, and processes. To imagine how this additional step could have come about, consider the following. The occurrence of feelings, no matter how modest, *coinciding with* or even *caused by* the presence of varied objects, actions, and processes, allows those constituents to be referred to their respective body and thus become "known" to their respective organism. In other words, *consciousness proper* begins with genuinely informative homeostatic feelings; such feelings – as well as others generated by standard emotive processes – are then naturally and spontaneously "applied" to and "associated" with other objects, actions, and processes. As a consequence, they too become connected to "their" organisms, which is another way of saying that they become "known" to the organism or that the organism is "conscious" of their presence.

The third step was the expansion of the *cognitive* machinery, an expansion responsible for refined imagetic perception of the world surrounding the organism as well as for the emergence of imagetic flow and of narrative making, both assisted by memory and reasoning. Intriguingly, this third step also amplified the organism's ability to experience the "world within," and expand the reaches of the feeling process itself.

In conclusion, we need to consider not one step but three, operating cooperatively but with distinct contributions. *Feeling* (the first step), which was primarily a motivator and guide of homeostatic behavior, enabled the second step – *consciousness proper* – by allowing the body reference that is integral to feelings to be applied to numerous other objects, actions, and processes. But *expanded cognition* (the third step) took advantage of the organism reference and promoted and perfected responses and permitted the invention of new solutions. These creative results accounted for the elaborate social behaviors which came to serve as the roots of cultural instruments and practices in the full sense of the term culture.

In summary, and in essence, the roots of cultural phenomena are to be found in the process of life regulation, that is, in *homeostasis*. Eventually, along with the development of *affective* processes and of their central feature – *feelings* – there arises a required expansion/complement: *consciousness proper*. The primary role of feelings and the indispensable presence of specific organism references are the critical features that allow homeostatic needs to motivate and guide the production of cultures. But the ultimate implementation of cultures also depends on the cognitive capabilities to which we allude above.

All of these stepwise processes – feelings, consciousness, and an expanded cognitive capacity – were in place before humanity first appeared. The entire range of cultural behaviors, the ones that emerged non-consciously along with the ones that were deliberately constructed by conscious, cognitively rich, creative minds, were the consequence of homeostatic needs. They still ARE. They follow the natural tendency of living organisms to respond to homeostatic needs efficiently and thus maintain life.

Eventually, when humans constructed collections of moral behaviors and religious beliefs, invented justice systems, practiced politics and economics, developed medicine, science, and technology, and engaged in the arts, they were still attempting to promote homeostasis by relying on (a) feelings, (b) the consciousness that they partly enable, and (c) a cognitive apparatus where imagination, memory, reasoning, and ultimately language had expanded spectacularly the possibilities of creative response.

1.2 From Life Regulation to Feeling

It is manifestly impossible to reconstitute the early evolutionary trajectory that begins with the processes of life regulation and ends with the emergence of feelings. Evolution cannot be described by a single line; there are several paths, parallel and intersecting, and different results, not to mention dead ends. It is reasonable, however, to point to probable stages and advance hypotheses about the capabilities of varied organisms along their evolutionary path, given the design of those organisms. When we consider unicellular organisms, with or without a nucleus, we simultaneously confront very simple and yet very complex creatures. There are rich processes of life contained within the perimeter defined by the cell membrane. Outside of that membrane there is the universe of other organisms and non-living objects; inside, there is an organized process of life, dependent on several organelles, each of which hails from even simpler life forms.

The machinery of life is through and through a world of chemistry, vulnerability, and transience. It is a world of atoms – those that are listed in the Periodic Table of Mendelev – combined and recombined in varied admissible arrangements so as to form molecules. The intelligence of this world has a specific but unstated aim – preserving the life process that consists of transforming energy, via metabolism, so that, for example, movements can be produced in response to chemical, mechanical, or thermal stimuli, arriving at the cell membrane. This intelligence is naturally embedded in the primary physical structure but modified by secondary chemical combinations, the resulting molecules, and their functional consequences. The life situation differs from the world of pure physics that undergirds the artificial intelligence of robots. It is vulnerable, even fragile, and it relies on continuous labors in order to stave off decline and extinction (Crick, 1981; Damasio & Damasio, 2021; De Duve, 1995, 2005; Dyson, 1999; Gantí, 2003; Man & Damasio, 2019; Schrodinger, 1944).

The term homeostasis refers to a collection of processes aimed at maintaining the

operations of life within parameters compatible with its continuation (e.g., pH, temperature, presence, and amount of certain molecules). There is no single "optimal" or "best" life state. There are several states that are conducive to life marching on and several states that announce the opposite. At both ends of the range lies dysfunction, disease, and death. When decline or danger are *internally and non-consciously sensed*, a number of corrections are automatically instituted aimed at maintaining life. In the vast majority of creatures that ever have lived, this effective rescue operation does not require consciousness or reason. This qualification continues to apply to the present moment: for most creatures alive today the saving corrections of homeostasis are engaged unknowingly and only unknowingly. Matters are remarkably different when we transpose the micro-level situation to the level of humans: *when life operations become problematic humans feel malaise; if the problem persists, we feel sick; and if nothing is done to solve the problem we die. When sources of energy are lacking, we feel hunger, when hydration dips we are thirsty; cut our airflow and we feel "air hunger."* Intriguingly, at the other end of the behavioral range, exorbitant attempts at homeostatic correction can result in manic activity and also announce sickness and death. More often than not, in one way or another, humans try to respond to the felt warnings and take advantage of them. In other words, humans have dual homeostatic controls, the early automated version and the newer feeling variety further elaborated on by consciousness proper (Bernard, 1879; Berridge & Kringelbach, 2015; Cannon, 1929, 1932; Craig 2002, 2003, 2009; Damasio & Carvalho, 2013; Damasio & Damasio, 2021; Richter, 1943; Torday, 2015).

It is imperative and helpful, once we introduce an illustration that comes from the human level, to add that the likelihood of anything comparable to explicit feeling or explicit cognition in unicellular organisms is quite low. We know, today, a little bit about the physiological complexity required for mental phenomena such as ours – for example, the ability to make maps on the basis of which imagetic representations can arise. One can make a partial exception for primordial feelings, where the imagetic component would be minimal and where the action of chemical molecules would dominate. But robust and in all probability specific feelings also require neural structures. Mental phenomena with any degree of specificity are difficult to conceive when we consider that mental phenomena would have to be generated from chemical pathways within the cytoplasm of a single cell, even when we grant that cell a number of differentiated processes and devices – a nucleus and organelles such as mitochondria. What such processes and devices do is remarkable, no doubt, but they have their hands full running a risk-prone chemical factory. Thinking is not likely to result from that primary assignment.

What we are claiming here is that the level of complexity of unicellular organisms is not suggestive of the presence of mind processes. Why not? We venture (and wager) that there are no images to be found in single cells, alone or combined in a narrative flow; that there are no devices or processes capable of constructing representations of themselves and of their "parts," let alone representations of the world exterior to the cell. We believe that there are, instead, complex chemical states, maintained within certain bounds; that there is production of energy which can be channeled to movement and permits responses to stimuli or the spontaneous initiation of actions, provided the responses and actions are helpful, in some way, to the continuation of life in that organism. At any moment, continuously, the cell exhibits a particular chemical "state of life"

that can be closer or further away from the most useful operational ranges for that particular organism. The part of the range toward which "states of life" tend to gravitate naturally defines the success or failure of the ongoing state of life. That degree of success or failure can express itself by certain overall actions of a cell – for example, relaxing its membrane, quietening its activities; or, on the contrary, by agitating its motions, constricting its perimeter and literally shrinking. Importantly, however, in simple organisms, those active expressions would NOT have a prominent mental counterpart. There would be no pronounced "mental state" to accompany the states conducive to life's continuation within a cell or to its demise. We suggest that, if single cells could have had *a mental expression of the state of life within their organism*, then that expression would have been a *feeling*, a natural translation of the goodness or badness of the state of life relative to its intrinsic goal: persist, maintain itself.

Our current position is that the entire sequence of processes – from basic life state to the mental expression of that state, a feeling – becomes possible only in organisms equipped with nervous systems of a certain degree of complexity. Still, we are open to the possibility that a number of species without nervous systems but with considerable multisystem differentiation might conceivably generate early stages of a feeling-like process and produce a basic sense of existence, a sense of being.

For the moment we need to face the reality that dating the beginning of feelings and connecting that beginning to specific species is not satisfactorily established. We need to accept that there is a gray zone, conceivably occupied by multicellular organisms with internal organ differentiation and with or without simple nervous systems, in which a general "sense of being" might be possible.

Beyond the gray zone, however, it is reasonable to imagine multicellular organisms with abundant internal organ differentiation and progressively more complex nervous systems with elaborate central components. And it is within such organisms that we expect feelings to have blossomed and have been capable of providing both a sense of being *and* an early manifestation of organism-based perspective on which consciousness can be based. In such organisms it is possible to conceive a broad range of homeostatic feelings thanks to the influence of circulating molecules on the operation of internal organs as well as on neural networks. Such molecules naturally help produce "valence," the one-word synonym for the *quality* of the life state relative to its fundamental goal: persisting.

1.3 The Physiology of Feelings

Given the significance of homeostasis and of feeling for the generation of social behaviors and, eventually, of cultures, it is important to address our current understanding of the physiology behind the emergence of feelings.

Nervous systems first appeared around 500 million years ago. As suggested in Section 1.2, although we are open to the possibility of primordial feelings emerging even earlier, we favor the hypothesis that full-fledged feelings require the presence of neural structures and fairly complex ones at that. The simpler variety of feelings would depend upon the action of chemical molecules. The configuration of the neural circuitry as determined by a particular chemical environment would signify and, in a way, "represent" a certain life state, a particular region or position along the homeostatic range. But the intriguing and often neglected issue as one attempts to describe this process is that, in this case, the "representation" does not constitute a conventional "perception" of a certain body state.

"Neural representation" and "actual body state" work in integrated fashion to be "felt" rather than simply perceived. They "interact." This distinction also points unceremoniously to the core of the feeling problem. We, the subjects of feeling, do not merely inspect "representations" or "perceptions" of our internal states telescoped by our senses to our brains. We, the subjects of feeling, are directly privy to such states and the term feeling designates that intimate process. The "perceiver/experiencer" *of a certain state of life within an organism turns out to be that same organism*. At first glance, the statement appears paradoxical and even absurd, but it turns out to be, upon reflection, the source of an explanation for the distinction between what is simply "perceived" and what is "felt" (Carvalho & Damasio, 2021).

In the typical situation of perceiving objects or events in the world external to an organism, there is a clear distinction and distance between the perceptual organ and the object of perception. This is not so in the world of feeling where there is no distance between perceiver and object of perception. There is, instead, an integration and fusion of the two participants, subject and object, such that they become difficult to tease apart. We believe that the difficulty in separating perceiver and perceived is part and parcel of the mechanisms behind the ability to feel.

When the retina (and, subsequently, the visual cortices) map a certain object in the surround of the organism, retina and object exist in separate realms, one inside the organism, the other outside. The perceived object cannot intervene in the process of perception directly, nor interfere with the chain of physiological changes that are occurring in different layers of the retina and in the varied cortical regions toward which the retina projects its signals. The same applies to the sounds processed in the cochlea relative to the auditory cortices or to the shape of objects touched by the skin of one's fingers. But when we are said to perceive the state of contraction of an internal organ such as the gut or perceive the consequences of opioid molecules circulating in our organism, the "perceiver" and what is being "perceived" *inhabit the same physiological space and can interact physically within the process. This is a fundamental distinction between the typical process of interoception (part of which is also known as visceroception) and the processes of exteroception*.

Exteroceptive processes such as seeing, hearing, and touching operate differently from interoception and even from proprioception (the perceptual system that refers to striated muscles and voluntary movement). Exteroception relies on perceptual maps which plot spatial patterns in one or more dimensions. Interoception begins to operate at the level of chemical molecules involved in altering the actual function and shape of viscera directly or via intermediate neural activity. At more complex structural levels, interoception also makes use of maps. For example, when we feel a contraction of the gut and call it a "colic," or the reduction of caliber of the trachea that hallmarks asthma. In such situations, feelings conform more closely to the notion of perception, but the process is still distinct from typical exteroception for obvious reasons. To a considerable extent, when it comes to feeling, the "subject" and the "object" merge, at least in part, and become less distinct. When we talk of "perception" under such circumstances we must add that the circumstances are unusual and, in a way, paradoxical. Helena De Preester has devoted particular effort to illuminating this problem (De Preester, 2019; Henry, 1963).

In conclusion, in a curious way, we might say that we generally do not need to "perceive" the physiological changes that underly our feelings because we *feel* them instead. Feelings

offer themselves to our mental experience, *directly*. We believe that this is due to the fact that the relationship between the neural devices that probe our life state and those same life states is of *close intimacy and oneness*.

Recent results are helping us glean the physiological features and processes behind the intimacy that result in feeling. Some of them concern fundamental peculiarities of the neural hardware related to affect (Carvalho & Damasio, 2019, 2021; Carvalho et al., 2019). The most important among them has to do with the kind of neurons involved in the physiology of feeling. Those neurons turn out to be predominantly non-myelinated. This amounts to depriving their axons of the insulation material – the myelin – that separates the axon cable from its chemical surround. For practical purposes this feature opens the axons of the respective neurons to the influence of varied chemical molecules present in their environment. The nervous system and the surrounding non-nervous organism structures partake of intimate chemical interactions. As a result, in addition to the neuron signaling that depends on the activity or quiescence of synapses, nature adds to the mix another kind of signaling – known as Non-Synaptic-Signaling – along with an infinity of graded states beyond simply "on" and "off."

One dramatic consequence of such circumstances occurs in nerve structures whose axons are aligned in parallel, in a sort of palisade, as is the case with the vagus nerve, a nerve which is crucially involved in feeling processes. As a result of the lack of myelin in the vagus nerve, it is possible for electrical impulses to be conducted sideways, orthogonally to the direction of the nerve, such that conduction spreads *across* the bunched axons rather than along them. This phenomenon is known as *ephapsis*, and it enriches the menu of neural communication to include not just synaptic but also ephaptic conduction.

We must note yet another contributor to the intimacy of interoceptive processes: the fact that certain structures of the interoceptive system lack a blood-brain-barrier. This is manifestly the case in the neural ganglia that bring signals from everywhere in the organism's soma to the central nervous system, namely, the spinal ganglia distributed up and down the spinal cord, and the trigeminal ganglia in the brainstem. There is no blood-brain-barrier in part of the ganglia, and in its absence the neurons become open to the direct influence of molecules circulating in the blood capillaries and in cerebrospinal fluid. They are in the same chemical bath. This is perhaps the closest degree of intimacy between "body" and nervous system.

In brief, thanks to the expression of homeostatic states in the process of feeling, our cognitive processes can be informed and advised of the momentary life conditions of our organism, thus becoming conscious and acting accordingly in a large variety of situations, including our social human experiences. It should be apparent that both the functional "rationale" for consciousness and the "problem" of explaining its physiology, are approachable through the avenue of feeling.

1.4 The Role of Emotions

In our attempt to provide a neuro-biological account of the development of cultures we have so far evoked feelings, consciousness, and cognitive processes. Clearly, we give pride of place to affect but we have not yet mentioned the phenomena that are most frequently, and incorrectly, invoked as the virtual emblem of affect: *emotions*. Where do emotions fit in our account? What are emotions and where do they stand relative to feelings? These definitions and relationships have recently become more approachable. So here

are the critical distinctions (Damasio, 2018, chapters 7 and 8, 2021).

First, as we have seen, feelings are through and through experiences of life states, although, as we have just shown, they form a distinct class of experiences.

Second, the fundamental variety of feelings is the one we have been highlighting: *homeostatic*. These feelings correspond to life states such as *well-being*, *malaise*, *pain*, *hunger*, *thirst*, and *desire*.

Third, emotions (and to a great extent drives and motivations) are *highly established*, *easily repeatable*, *largely involuntary action-programs*. Some of the actions in the program are only internal (contraction of smooth muscles of viscera, secretion of certain molecules), and some can be observed externally (relaxation or contraction of striated muscles, specific patterns of facial expression or gait). We can observe emotions in other creatures, of course, but we can also *feel* them in ourselves when we experience them as *emotional feelings*.

Fourth, *emotional feelings are transient modifications of the ongoing homeostatic state, large and small, caused by the deployment of emotional action-programs*. In brief, emotional feelings are transient variations (one might also call them "disturbances") of the ever-flowing homeostatic states. Consequently, emotional feelings *constitute variations on the theme of homeostatic feelings*.

Fifth, unlike homeostatic states and their respective feelings, which are part and parcel of the essential life process, emotions are *elaborate concerts of actions* that evolved to address and cope with specific problems faced by organisms, including problems posed by life in a social setting. That is why emotion programs prominently include fear, anger, joy, sadness, compassion, love, pride, humility, disgust, and contempt. To a greater or smaller degree, all of them constitute attempts to resolve, in a prompt, unreasoned manner, a problem posed by life, especially in *society*.

Emotions are "triggered" by the ongoing circumstances of life. The presence and perception of particular objects or events "engages" emotional responses which result in subsequent emotional feelings.

The flowing of events in the narratives of our lives is replete with emotive reactions and their attendant feelings, thus irrevocably intermixing cognition and affect.

We have discussed the role of affect in cultural life using the notion of "somatic marker" in domains as diverse as ethics, esthetics, economics, and politics. (See, for example, Damasio's *The Strange Order of Things* 2018; and Verweij & Damasio, 2019.)

1.5 Building Cultures

We have outlined three steps responsible for the building of complex social behaviors and, eventually, of cultures in the full sense. The three steps are: (1) *feelings*, the translation of homeostatic states into *hybrid somatic and mental experiences*; (2) *consciousness proper*, the largely feeling-derived functional arrangement that refers mental representations to an ongoing life process unfolding in a *specific organism*; and (3) the expansion of *cognitive* abilities responsible for translating in imagetic terms the presence of objects, actions, consequences, and qualities, actually occurring in the world external to the organism or imagined on the basis of reasoning over images, freshly minted or recalled from memories. The flow of images in a conscious mental process is well designated by the term *narrative*. Narratives are the fundamental currency of mental life, responsible simultaneously for telling the story of what things are and of how they interact in time but also capable of telling us what things can become or even ought to be (Kaplan et al., 2016).

Plain narratives operate on the basis of imagetic representations, spatial patterns more or less faithful to the originals that they stand for, often schematically simplified to the point of abstraction or translated in the form of symbols. The most commonly used set of symbols belongs to verbal languages. It is quite common for narratives to make use of both direct imagetic representations as well as linguistic translations.

One of the intriguing aspects of narratives is that they act upon organisms by using not only cognitive processes – reasoning, recall of previously memorized material, creation of new compositions – but also by prompting all manner of *affective reactions*, namely, *drives*, *motivations*, and *emotions*. As noted in Section 1.4, as these affective reactions are organized and deployed, organisms apprehend their significance and consequences via *emotive feelings*. The process sounds complicated at first but really is not. It is best conceived when one realizes that as life flows with all its attending vagaries, the introduction in its midst of an emotion or drive operates primarily by having the sets of actions that constitute, say, an emotion *disturb* the ongoing life process. The disturbance is such that, for example, certain visceral parameters are temporarily altered – the typical examples being cardiac and respiratory rhythms; gut activity; and myriad endocrine reactions. In other words, the ongoing cognitive narrative is responsible for prompting affective responses; such responses, for example, emotions, alter the flow of life so that the contents of the flowing narrative are not merely *accompanied* by affective phenomena – in a parallel track – but actually *modified and accompanied* by those phenomena. Moreover, in a wave of subsequent reciprocation, the new contents of the now enriched cognitive narrative can generate their own emotive responses.

The consequences of this arrangement are manifold. First, feelings introduce within the mental process information regarding the state of life in the organism and make that information count if it happens to be relevant. One can say that feelings plant, within the cognitive, creative process, the latest bulletins from the homeostasis front. Second, the idea that cognitive processes might operate independently from the world of feeling is pure fantasy and the same applies to the idea that the processes of life regulation and cognition would operate independently of each other.

The link between life regulation/homeostasis and the creation of cultures hinges on this curious physiological arrangement.

1.6 In Closing

In this essay we have tried to demonstrate the integral connection between biological phenomena and the development of cultures. We are following sound historical precedents. The moral philosophy of Adam Smith was grounded on understanding human affairs from a biological perspective. Intriguingly, Charles Darwin was influenced by Adam Smith as he applied selection mechanisms to cultural phenomena in *The Descent of Man*, as Marciano and Pelissier (2000) have demonstrated. But the close connection between Darwin and Smith reminds us that both were well aware of the precarity of life and of the extraordinary machinations behind its regulation at nonconscious as well as conscious levels.

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2 Epigenesis, Synapse Selection, Cultural Imprints, and Human Brain Development

From Molecules to Cognition

Jean-Pierre Changeux

2.1 Introduction: The Singularity of Brain Organization and Synaptic Epigenesis

The human brain is neither John Locke's "blank slate" deprived of any pre-existing innate structure – or, in a modern AI language, a random network of undifferentiated neurons fully instructed by experience – nor a fully genetically determined, irrevocably hard-wired neuronal architecture. Neither is it represented by the simplistic yet very popular deep-learning artificial networks. The 85–100 billion neurons of the human brain and their synaptic connections, that arose over million years of evolution and for each individual brain over almost 15 years of postnatal development, possess an original organization unmatched by any of our current computers. It is a unique compromise between an eminently variable, intrinsically rich connectivity and a set of species-specific, genetically determined, rules, which unambiguously make our brain that of *Homo sapiens*.

From a systemic point of view, the brain may be described as a dynamic assembly of multiple functional and reciprocally inter-regulated levels of structural organization, bottom-up, from molecule to consciousness and, conversely, top-down from cognitive functions to molecular processes, unified in a single global system. Actually, the brain may be viewed as the seat of multiple nested spontaneous evolutions operating concomitantly through variation–selection–amplification ("Darwinian") mechanisms (Barkow et al., 1992; Campbell, 2016; Changeux, 1983, 1985; Changeux & Danchin, 1976; Changeux et al., 1973; Edelman, 1987, 2006; Morange et al., 2016). These evolutions develop on strikingly different time scales: from the million years of the descent of man up to the 100 ms of psychological processes. They further operate from diverse variable units distributed in parallel at multiple nested levels of physical organization through variation–selection–amplification mechanisms creating profound intertwining between the developing (and adult) neuronal organization of the brain and the constantly evolving physical, social, and cultural environment. The brain's morphogenesis is progressive, with forms becoming intricated within forms, including possibly, at each step, sensitive phases of interactions with the environment. This dual relationship between the developing brain and its environment introduces an essential variability in the anatomy and functional architecture of the brain, which makes any individual human brain unique from both a

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connectional and a behavioral point of view (Changeux, 2017).

The aim of this chapter is to present and discuss a theory – initially expressed as a mathematical model (Changeux et al., 1973) – that gives access to such "epigenetic" inscription of environmental features within the developing connectivity of the brain. These views were consistent at the time with the behavioral observations of the newborn "learning by unlearning" (Mehler 1982) and with the then available knowledge of molecular biology. They did not need to refer to any sort of "instructive" or so-called Lamarckian mechanism (Piaget 1976; Quartz & Sejnowski, 1997). Furthermore, the term "epigenesis" (or "epigenetics") was used in a sense close to its original definition by Waddington (1942) and to his concept of the "epigenetic landscape" to illustrate how external events, some random, combine with inherited information coded in the genes to produce acquired variability between individuals from the same species. A well-established case is the acquisition of oral and written language, which, as discussed in Section 2.7, leaves important connectomic traces in the brain, which are re-learned, from one generation to the next.

This meaning differs from the concept of DNA "epigenetics" subsequently used in molecular biology to refer to unrelated mechanisms of DNA covalent modifications such as methylation and/or chromatin remodeling, which contribute to the regulation of genes expression without altering the DNA sequence (Lucchesi, 2019). The long-term heritability of such phenotypic changes has been suggested but heavily questioned. In addition, DNA "epigenetics" deals with gene expression at the level of the single nucleus of the nerve cell, while each individual neuron may establish up to 100,000 synapses available for synaptic epigenesis. Several levels of brain organization thus distinguish restricted DNA "epigenetics" from a broader synaptic epigenesis.

In the brain, the macromolecular level holds a decisive role by imposing inescapable bottom up physical constraints upon higher – up to the highest – functions (Changeux, 1983, 1985, 2017). First of all, the evolution of the species-specific features of the brain is grounded in the genes of the organism – in particular those which determine the "proto-organization" (Arcaro & Livingstone, 2017; Geschwind & Rakic, 2013; Rakic, 1976) or scaffolding of human brain connectivity – that we have referred to as the "genetic envelope" (Changeux et al., 1973). Yet, the issue of the evolution of the genetic envelope compared to the brain phenotype raises an interesting paradox. From mouse to humans, the size of the brain and, in parallel, its total number of neurons, increases from approximately 40–70 million to 85 billion (plus 50 billion glial cells), and the number of specialized cortical areas per hemisphere rises from about 10 in primitive mammals to as many as 180 in humans (Markov et al., 2013). In addition, the intrinsic connectivity, especially in the cerebral cortex, dramatically increases in diversity, leading to the ultimate development of oral language in less than 1 million years (Goulas et al., 2019).

By contrast, the full genome sequences now available for many species – mouse, monkey, chimpanzee, humans, and fossil human ancestors (Paabo, 2013) – are striking in their relative uniformity. The haploid genome comprises no more than about 20,000 gene coding sequences (only 1.2 percent of the human genome) and this number does not vary significantly from mouse to humans (Geschwind & Rakic, 2013; Paabo, 2013; Somel et al., 2013; Vallender et al., 2006). The coding genome itself appears highly conserved, especially for brain proteins, even at

the most recent stages of hominization. The available comparative genomic data unambiguously reveal that the observed increase of brain anatomical and functional complexity does not reflect a parallel increase in the complexity of the genome (Dumas et al., 2019). This may be seen as an "astonishing evolutionary parsimony" (Changeux 1983, 1985, 2017). Still, at this stage, the actual few discrete genetic regulatory events that determined the fast increase in brain complexity during the past million years of hominization remain unidentified. These might include discrete changes in genomic organization, including gene duplications (Suzuki et al., 2018), DNA regulatory sequences (Petr et al., 2019; Weyer & Paabo, 2016), and others concerning the relationships between genes and phenotypes that have been under-evaluated within the classical "one gene–one protein–one phenotype" paradigm. The selective stabilization hypothesis is here presented as one mechanism, which might contribute to this evolutionary parsimony (Changeux 1983, 2017).

2.2 The Model of Changeux–Courrège–Danchin (CCD), 1973

During embryonic and postnatal development, the million billion synapses that form the adult human brain network do not assemble, like the parts of a computer, according to a plan that defines with great precision the disposition of all individual components. If this were the case, the slightest error in the instructions for carrying out this program could have catastrophic consequences. The mechanism proposed, on the contrary, relies upon the variability of developing interneuronal connections (within the mentioned genetic envelope) and the progressive setting of robust synapses through trial-and-error mechanisms that formally resemble an evolutionary process by variation–selection (Arcaro & Livingstone, 2017; Blakemore et al., 1981; Changeux, 2017; Changeux & Danchin, 1976; Changeux et al., 1973; Edelman, 1978; Kasthuri & Lichtman, 2003; Purves & Lichtman, 1980; Rakic, 1976; Rakic et al., 1986; Shatz & Stryker, 1978; Sretavan et al., 1988). A most important and unique feature of human brain evolution is the extension of postnatal development for up to fifteen years. An approximately five-fold increase in brain weight accompanies this development, during which about half of all adult synaptic connections are formed at a very fast pace (approximately 0.5 million synapses per second) (Lagercrantz et al., 2010). The model was essentially designed to apply for the postnatal period of synaptogenesis of the human brain but may also account for some earlier steps of prenatal synaptogenesis and for the adult brain, yet to a lower extent (Petanjek et al., 2011).

On formal grounds, the original concept (CCD model, Changeux et al., 1973) has been extended with the "neural Darwinism" (Edelman, 1978), the group-selection theory of higher brain functions (Edelman, 1981), and more recently with the theory of symmetry breaking in space-time hierarchies (Pillai & Jirsa, 2017), among others.

The CCD model is based on the principle that interneuronal contacts, the synapses, mediate information transfers through the system (Figure 2.1). At nested critical periods during the development of the brain, the phenotypic variability of nerve cell distribution and position, as well as the exuberant spread and the multiple transient connectivity configurations resulting from the growth cone wanderings, produce a broad diversity of synaptic connections. This transient diversity is then reduced by the selective stabilization of some of the labile contacts and elimination (or retraction and/or pruning) of others. Excitatory as well as inhibitory synapses may



Figure 2.1 The neurone and the interneuronal connections through the synapse in the human brain
A black-and-white version of this figure will appear in some formats. For the color version, please refer to the plate section.

exist under at least three connective states: Labile (L), Stable (S), and Degenerate (D); only states L and S transmit nerve impulses and the acceptable transitions between states are L→S, L→D, and S→L. A critical implication of the model is that evolution of the connective state of individual synaptic contacts is governed globally, and within a given time window, by the total activity afferent onto the postsynaptic soma during a prior time interval of determinate length (evolutive power of the soma). It includes, as a particular case, the standard Hebbian time-coincidence relationship and the popular statement "cells that fire together wire together." Activity of the postsynaptic cell in turn regulates, in a retrograde manner, the stabilization and/or elimination of afferent synapses.

The maximum wiring and the main stages of development of the network of synaptic connections, as well as the evolutive power and the integrative power (after the usual "firing" mechanism) of each soma are determinate expressions of the genetic endowment (the "genetic envelope" of the network). The emergence during growth of a large number of labile synapses is provided by this species-specific envelope. The associative property that results from the "learning process" is structurally printed as a particular pattern of such organization. This pattern results often from the selection by functioning of particular pathways among a large number of labile synapses (especially during growth). Such trial-and-error mechanisms formally resemble the variation–selection process of an evolutionary Darwinian but epigenetic process (Campbell, 2016) (see Figure 2.2).

A nesting of many such elementary steps occurs in the course of development. The growth of axons toward their targets – the dendrites of target neurons in the central nervous system or muscle cells in the periphery – involves cell-surface recognition molecules, possibly ones unique to the specific category of connections. The axon terminals branch exuberantly at first. But, then, depending on the state of activity of the target neuron – either intrinsic spontaneous firing or evoked by external inputs – some synapses are eliminated (pruned), while others are strengthened and stabilized. In post-natal life, an important part of the activity in the network results from inputs from the environment and so the epigenetic selection of synapses represents learning in the network.

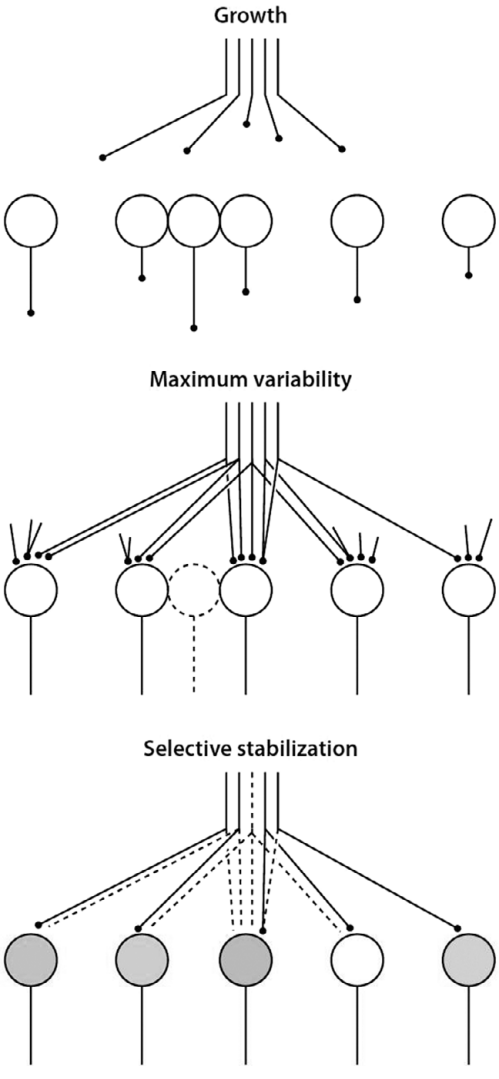


Figure 2.2 Schematic representation of the hypothesis of epigenesis by selective stabilization of synapses. From Changeux, 1983, 1985
A black-and-white version of this figure will appear in some formats. For the color version, please refer to the plate section.

2.3 The Variability Theorem and the "Multiple Realizability" of Brain Functions

The CCD model or theory accounts for the interactions that take place between the brain and its physical, social, and cultural environment in the course of cognitive development. But it accounts in addition for a property largely under-evaluated in brain research: the variability in the brain's connectivity and in behavior between individuals, which is associated with the variability of the environment and would be superimposed on that created by the variability of the genome.

As mentioned, in the course of the proposed epigenesis, diversification of neurons belonging to the same category occurs by the acquisition of the precise pattern of connections it establishes (and neurotransmitter/receptors it synthesizes) (Changeux, 1983, 1985). An unexpected but critical feature of the theory is that it may account for the constancy of some behaviors, despite such epigenetic variability of the connectivity. This idea was originally stated as the "variability theorem" (CCD model) that "different learning inputs may produce different connective organizations and neuronal functioning abilities, but the same behavioral abilities" (Changeux et al., 1973, p. 2976). Thus, the neuronal connectivity code exhibits "degeneracy" (see Edelman, 1978; Edelman & Gally, 2001; Tononi et al., 1999), that is, different code words (connection patterns) may carry the same meaning (function).

One prediction of the variability theorem is that the synaptic connectivity of genetically identical individuals (i.e., monozygotic twins) may display phenotypic variance. This was initially demonstrated using serial electron microscopy scans of genetically identical individuals (identical twins). In the small invertebrate Daphnia magna, the number of cells is fixed and the main categories of contacts (between optic sensory cells and optic ganglion neurons) are preserved from one isogenic individual to another. Yet, the exact number of synapses and the precise form of the axonal branches vary between pairs of identical twins

(Macagno, Lopresti, & Levinthal, 1973). Similar findings have been reported in the case of the Müller cells of a parthenogenetic fish (*Poecilia formosa*) (Levinthal, Macagno, & Levinthal, 1976), and thus they are not restricted to invertebrates. In a general manner, a phenotypic variance of the connectivity exists. This has been exquisitely demonstrated in the rat, where a detailed left–right comparison of interscutularis muscle innervation by the same motor neurons (Lu et al., 2009) revealed a profound variability of axon branching. This has been confirmed with the nervous system of isogenic Caenorhabditis, which was until recently viewed as precisely wired up to the individual synapse level (Oren-Suissa et al., 2016). Yet, the Daphnia, the worm, the fish, and individuals all swim similarly, and the left and right limbs of the rat also work the same!

In mammals, the variation also affects the number of neurons. For instance, in the case of the cerebellum of the mouse, the division and migration of Purkinje cells in consanguineous strains are not subject to as rigorous and precise a determinism as the laying down of neurons in invertebrates (Goldowitz & Mullen, 1982; Oster-Granite & Gearhart, 1981). The variability becomes microscopic and may even affect its chemistry, such as the type of neurotransmitter through activity dependent switches (Spitzer, 2017) or the pattern of transmitters and coexisting messengers synthesized (Hokfelt et al., 1986, 2018) of entire populations of neurons.

At a much higher level of complexity, humans with language areas located either in the left or right hemispheres, or in both, are indistinguishable by the way they speak or think! In monozygotic twins, differences in manual preferences have even been discovered, which by itself is a sign of variance at the behavioral level. In vivo measurements using magnetic resonance imaging of the planum temporale further reveal brain images that differ between left- and right-handed twins, which is more pronounced in the right-handed one (Sommer et al., 2002; Steinmetz et al., 1995).

In humans, most of the information available on anatomy derives from individuals taken from genetically heterogeneous populations. Nevertheless, even under such conditions, the substantial variability noticed, and, thus, the development of the fine details of the brain connectivity pattern, is expected to include a stochastic element. Chance plays a part in determining exactly which synapses survive. Nevertheless, the behavior may remain constant between individuals. These views are consistent with recent formal descriptions of brain behavior as a low-dimensional process emerging from a network's dynamics depending on the symmetry and invariance properties of the network connectivity (Pillai & Jirsa, 2017). These authors show that given behavioral patterns can be accomplished through the use of distinct architectures utilizing (at least partly) different functional modes (engaging the same or different neuro-muscular linkages) (Perdikis et al., 2011). They conclude that since the different control structures are likely to draw on distinct neural network components, this prediction speaks to the concept of degeneracy, which is precisely what we proposed as "variability" in the early formulation of the CCD theory (Changeux et al., 1973). There is no contradiction between epigenetic "Darwinian" selection and the occurrence of behavioral universals. The synapse selection model offers a neural example to "multiple realizability" – that is, the non-unique (degenerate) mapping of a given "invariant" function to the underlying neural organization (Changeux et al., 1973; Pillai & Jirsa, 2017). This let some cognitivist philosophers incorrectly "believe" that, in particular because of this degenerate mapping, the

attempts to define a precise relationship between the connectional organization of the brain and its cognitive – in particular language – abilities is either irrelevant (Fodor, 1983) or even might remain a "mystery" forever (see Chomsky, in Hauser et al., 2014).

2.4 Cellular and Molecular Mechanisms of Synapse Selection

Both the spontaneous and the evoked activity may contribute to the synapse selection process. In this framework, the suggestion was made that reward signals received from the environment may control the developmental evolution of connectivity (Hull, 1943; Skinner, 1981; Thorndike, 1911; see also Gisiger & Kerszberg, 2006; Gisiger et al., 2005). Positive reward is signaled by neurons in the brain stem that release dopamine in the frontal cortex, whereas serotonin neurons signal negative reward, or punishment (Dehaene & Changeux, 1991, 2000; Kobayashi & Schultz, 2014; Stauffer et al., 2016). The evolution of connectivity through selection has been tested using a network simulation that can learn to do specific tasks when given simple positive and negative rewards (Gisiger et al., 2005). Before learning, the connectivity in the network is largely diffuse and unstructured and task completion is unsuccessful. After learning, the selected connections form a coherent and organized network that can complete tasks successfully. This is illustrated in the case of a visual delayed-matching-to-sample (Gisiger et al., 2005) or in a logical reasoning task (Houdé et al., 2000), both involving a prefrontal network. Further work should establish the actual contribution of reward to synapse selection in the course of development.

The cellular and molecular mechanisms involved in synapse selection are rather diverse. They include GABAergic inhibition, which contributes to the "opening" of the critical period where synapse selection occurs (Hensch, 2005; Werker & Hensch, 2015). A shift in the excitatory–inhibitory balance is associated with the maturation of fast-spiking GABAergic inhibitory neurons that synthesize parvalbumin and are localized in layers III/IV of the cerebral cortex (Takesian & Hensch, 2013). Pharmacological agents that accelerate GABAergic circuit function (such as benzodiazepines acting as positive allosteric modulators of $GABA_A$ receptors) elicit precocious onset, whereas genetic manipulations (such as the deletion of genes involved in GABA synthesis) or environmental disruption (such as dark rearing or hearing loss) lead to a delay of the critical period.

During early postnatal development, microglia undergo morphological maturation that matches synaptic maturation and possess phagocytic properties, which led to the hypothesis that microglia may have a role in the elimination of exuberant synaptic connections during development. This hypothesis is supported by studies that reported the appearance of excess immature synapses in mice lacking either the fractalkine (Cx3cl1/Cx3cr1) or complement component (C1q/C3/CR3) microglia signaling pathways (Paolicelli et al., 2011; Weinhard et al., 2018).

Following gene expression in the course of synapse selection in the visual system, it was found that among the genes regulated by neural activity are the MHC (major histocompatibility) Class I family genes (Corriveau et al., 1998). This finding was rather unexpected because these genes – HLA genes in humans – are involved in cellular immunity and were previously not thought to be expressed by neurons. Other components of a signaling system for Class I MHC are also present in neurons, including their receptor, PirB (Kim et al., 2013). The study of mice deleted for theses diverse molecular

components revealed that they are required for activity-dependent synapse selection (Huh et al., 2000). Homologs are present in the human brain and may have some relevance for understanding brain wiring developmental disorders including Alzheimer's disease (Kim et al., 2013).

Neurotrophic factors, including nerve growth factor (NGF), brain-derived neurotrophic factor (BDNF), neurotrophin-4 (NT-4), and neurotrophin-3 (NT-3) (Mandolesi et al., 2005), are important regulators of visual cortical plasticity (Huang et al., 1999). In transgenic mice in which the postnatal rise in BDNF in the forebrain was genetically accelerated, a precocious termination of the critical period of ocular dominance plasticity was found, which correlated with an accelerated maturation of GABAergic inhibitory circuitry. Neurotrophic factors are important modulators of synaptic epigenesis and brain development in general.

In addition, homeoproteins, such as Otx2, have been reported to have a role in synaptic evolution during critical periods of development (Prochiantz & Di Nardo, 2015). For instance, Otx2 accumulates in an activity-dependent manner inside the fast-spiking GABAergic neurons. Intriguingly, Otx2 is not expressed by these cells but instead is imported through diffusion from one or several external sources (Spatazza et al., 2013). Therefore, it was proposed that, in the mouse, Otx2 accumulation by fast spiking neurons cells is necessary and sufficient for a binocular critical period opening at twenty days after birth and closing at forty days (Prochiantz & Di Nardo, 2015).

Other work has revealed detailed mechanisms involved in synapse selection at the molecular level. Activity-dependent changes in the diffusion dynamics of postsynaptic receptors present under the nerve terminals have been recorded even if the density of receptors remains stable under the nerve ending (Triller & Choquet, 2008). Also, allosteric transitions of the NMDA receptor traps the diffusible D1 dopamine receptor, resulting in an increase in dendritic spines bearing dopamine receptors (Scott et al., 2006). Such plasticity phenomena might possibly be altered in neuropsychiatric disorders.

Several molecular and cellular mechanisms are available for the active selection process to take place. They are themselves under stringent genetic control and belong to the genetic envelope of the species.

2.5 Anatomical and Physiological Evidence Supporting the Selective-Stabilization of Synapses Model

The classical Wiesel and Hubel experiments with the cat about the effects of monocular deprivation on binocular vision are consistent with the selective stabilization hypothesis (Hubel & Wiesel, 1965; LeVay et al., 1980; Shatz & Stryker, 1978; Sretavan & Stryker, 1988). If, during a critical period in early life, one eye is allowed to see normally, whereas the vision of the other eye is occluded, most of the cortical cells, even those in the deprived eye's columns, lose their ability to respond to the deprived eye (Hubel & Wiesel, 1970; Hubel et al., 1977; Shatz & Stryker, 1978; Wiesel & Hubel, 1963). This physiological loss of response is followed within one week by a dramatic retraction of the branches of deprived geniculocortical arbors and is later followed by a compensatory expansion of the arbors of the open eye (Antonini & Stryker, 1996; LeVay et al., 1980; Rakic, 1976). The effects of deprivation can be reversed to a limited extent during the critical period but they later become irreversible (Antonini & Stryker, 1998; Blakemore et al., 1981; Movshon, 1976; van Sluyters, 1978; Wiesel & Hubel, 1965). An example of such an irreversible effect of epigenetic experience in

humans is amblyopia, an eye disorder characterized by impaired vision in one eye that otherwise appears normal. It may be viewed as a "cortical blindness" that occurs during early childhood, as a consequence of cataract or strabysmus.

Early demonstrations of the process of synapse stabilization accompanied by synapse elimination of supernumerary nerve endings include work on motor neurons innervating skeletal muscle cells (Benoit & Changeux, 1975, 1978; Gouzé et al., 1983; Henderson et al., 1986; O'Brien et al., 1977; Rakic et al., 1986; Turney & Lichtman, 2012). In the adult, there is only one motor nerve ending per muscle fiber, but at birth several active motor nerve endings converge to a common endplate (Redfern, 1970). All of them except one are subsequently eliminated and the state of activity of the neuromuscular junction plays a decisive role in this elimination (Benoit & Changeux, 1975, 1978). This epigenetic step results in the formation of the motor units, which work together to coordinate the contraction of the whole muscle (Buchtal & Schmalbruch, 1980).

In the cerebellar cortex, similarly, the interactions between parallel fiber and climbing fiber synapses on Purkinje cells reveal heterosynaptic competition during development. The mature monoinnervation of Purkinje cells) by their climbing fiber, afferents from the inferior olivary nucleus is preceded by a transient stage of multiple innervation (Mariani & Changeux, 1980). In the rat, synapse exuberance reaches a maximum of up to five climbing fibers per Purkinje cell (about 3.5 on average) on postnatal day 5 and then regresses until monoinnervation is established on postnatal day 14–15. Granule cell precursors suppression by postnatal X-irradiation indicates that granular cells are not involved in the early part of climbing fiber synapse elimination but only the final phase, after postnatal day 8, would be driven by parallel fibers activity (Lohof et al., 1996). Genetic mutations have also been shown to alter synaptic epigenesis in the mouse (Changeux & Mikoshiba, 1978).

The innervation of the sympathetic submandibular ganglion cells undergoes profound reorganization during postnatal development, leaving individual ganglion cells in the adult innervated by many-fold fewer axons than they were at birth (Lichtman, 1977, 1980; Sheu et al., 2017).

These and many other studies (Bailly et al., 2018; Kano & Hashimoto, 2012; Ko et al., 2011; Li et al., 2017; Luo & O'Leary, 2005) have shown that, when neuronal activity is artificially modified, synaptic elimination is altered. At variance with the "Lamarckist-constructivist" scheme (Quartz & Sejnowski, 1997), blocking the activity maintains a high number of connections. Activity thus enhances synaptic elimination. As I have said in the past, "to learn is to eliminate" (Changeux, 1983, 1985). Many groups (Shatz, Lichtman, and others) have since abundantly documented this. Extending this idea to postnatal cognitive development itself (from object permanence in infants to logical reasoning in adolescents and adults), Olivier Houdé said, "to develop is to inhibit" (Houdé, 2000, 2019).

The classical information-processing scheme of the nervous system is based on the notion that its internal states of activity directly result from interactions with the outside world. In fact, from very early on, there is intense spontaneous electrical and chemical activity within the nervous system of the embryo and the fetus (Hamburger, 1970; Provine & Ripley, 1972). Chick embryos move within the egg as early as after two days of incubation. These spontaneous movements are blocked by curare (Levi Montalcini, 1936, see 2000) and coincide with electrical activity of the same frequency arising in spinal cord neurons. In the human, these movements start during the eighth week of

embryonic development and diversify during the following months. Such spontaneous activity develops in a strictly endogenous manner and results from molecular oscillators consisting of slow and fast ionic channels (Berridge & Rapp, 1979). The cost of this activity in terms of structural genes is very small and bears no relation to its enormous potential for interaction and combination, which results from the activity's eventual contribution to the epigenesis of neuronal synaptic networks. In the retina, the spontaneous activity of the ganglion cells (Galli-Resta & Maffei, 1988) forms correlated patterns, or waves (Maffei & Galli-Resta, 1990, confirmed by Meister et al., 1991; Shatz, 1996).

The mechanism of wave initiation and propagation is not fully understood, yet it involves synaptic transmission mediated by nicotinic acetylcholine receptors (nAChRs) (Feller et al., 1996; Penn et al., 1998). These waves of endogenous retinal activity are shown to play an important role in the epigenesis of neural networks through selective synapse stabilization. In mice deprived of the high-affinity nAChR beta2 subunit (Picciotto et al., 1995, 1998; Zoli et al., 1998), the pattern of retinogeniculate and retinocollicular projections was found to be altered (Rossi et al., 2001). In contrast, although alpha 4 subunit is the predominant partner of the beta 2 subunit in the formation of brain high affinity nAChRs, its deletion did not cause defects in retinofugal segregation. The initial developmental phase of retinal projections looked normal in beta 2 mice until postnatal day 4. This result, together with previous observations showing that the anatomy and the spatial resolution of the retina were normal in beta 2 mice, indicate that the deficit observed at later ages is not due to defects in retinal ganglion cells, programmed cell death or path finding errors of optical nerve fibers, but to the aberrant segregation of ipsilateral and contralateral axons at the target level. Consistent with these data, the beta-2 deleted mice show an expansion of the binocular subfield of the primary visual cortex and a decrease in visual acuity at the cortical level but not in the retina. The nAChR beta 2 subunit is thus necessary for the developmental epigenesis of the visual system at both anatomical and physiological levels (Grubb et al., 2003; Rossi et al., 2001).

The example of the visual system may plausibly be extended to other sensory motor systems and to a broader framework of a formal architecture of thalamocortical areas, in which top-down activity generated in hierarchically higher cortical areas plays a key role. Multiple nested "waves" of synaptic epigenesis take place during the postnatal development of the adult brain. As a consequence, a net decline of the total envelope of synaptic connections, which themselves sum up this laminated development, is observed (Huttenlocher & Dabholkar, 1997). In the primary visual cortex, for example, after a burst of synapse formation between the ages of three and four months, synaptic density reaches its peak at 140–150 percent of adult levels between the ages of four and twelve months, after which the mean number of synapses per neuron declines (Huttenlocher, 1990). The decline observed during late childhood plausibly reflects the underlying rich nesting of selection steps in a cascade of critical periods that proceeds far beyond puberty. In rats, the maximum synaptic density is reached within a few weeks after birth, whereas in humans it takes over three years (Bourgeois, 1997). Moreover, rats show little loss of synapses after maximum density is reached. On the other hand, in humans there is a steady decline until the total number stabilizes about the time of puberty (Bourgeois, 1997; Huttenlocher & Dabholkar, 1997; Petanjek et al., 2011), reflecting the initial exuberance and later

pruning of connections. In contemporary humans, the process of synaptic refinement goes far beyond puberty: learning is lifelong (Petanjek et al., 2011). When H. sapiens appeared in Africa about 100,000 years ago, half the average life span of around 30 years would have been taken up with building the brain. Cat and monkey show intermediate stages of this process as life span and infant dependency increases. This major distinction between lower mammals and humans has to be borne in mind to understand the evolutive process of brain humanization. This places severe limitations on using lower mammals as models for human psychiatric and neurological conditions.

2.6 Epigenesis of Higher Brain Functions and Global Integration

Among the cortical connections established in postnatal life are the long-range tracts between the frontal areas (Fuster, 2015) and other brain cortical areas (including sensory ones) (Collin & van den Heuvel, 2013; de Lange et al., 2019; Goldman-Rakic, 1988, 1999; Hagmann et al., 2008). Some years ago, the "global neuronal workspace" hypothesis was suggested, according to which these long-range connections, primarily between prefrontal, parieto-temporal and cingulate cortices, yield subjective conscious experience (Dehaene & Changeux, 2011; Dehaene et al., 1998) by broadcasting signals to multiple brain areas, thus allowing sensory inputs – such as seeing, hearing – conscious access to a whole brain "global workspace" (Baars, 1989).

Long-range connections, mostly originating from the pyramidal neurons in cortical layers II and III, are particularly abundant in the prefrontal cortex (von Economo & Koskinas, 1925), and form white matter bundles, several of which originate from the prefrontal areas (Dejerine, 1895; Pugliese et al., 2009). Particularly important are these connections involved in planning, decision making, thought, and socialization, which have evolved most dramatically between mice and humans (see Dehaene & Changeux 2011; Houdé, 2000, 2019).

The ontogeny and postnatal development of long-range connectivity expectedly reveal phases of exuberance and phases of selection and axonal pruning (Collin & van den Heuvel, 2013). Around birth, all major white matter tracts appear to be in place (Dubois et al., 2006; Hermoye et al., 2006), with at birth an over-representation of long-range connectivity followed rapidly by a decrease on the macroscopic scale. In nonhuman primates, a staggering 70 percent of callosal connections are eliminated (Innocenti & Price, 2005; LaMantia & Rakic, 1990). In human newborns, the evolution is slower. It has been suggested that the phase of exuberant axonal removal at the age of two years is completed accompanied by increasing information processing and cognitive – in particular language – development together with the "theory of mind" acquisition (Collin & van den Heuvel, 2013). The evolution continues during adolescence until adulthood with decreasing segregation and increasing integration mainly driven by modulation of connections stability and strength (Hagmann et al., 2010). It is expected to have major consequences on the laying down of cultural imprints, including the "epigenetic rules" associated with socialization.

One essential aspect of postnatal development in the human brain is the epigenesis of long-range connections establishment (Huttenlocher & Dabholkar, 1997) between different cortical areas, especially those of the GNW linking the parieto-temporal-cingulate cortices with the prefrontal lobes, which are the centers for decision-making, rational thinking, and social interaction. It is expected that these different areas develop rather

independently (as for instance the visual areas for each eye) but need a strong synapse selection step to integrate them into a unique global workspace at the scale of the individual (as an integrated binocular vision in the visual system).

One question currently asked is whether these long-range connections are especially vulnerable to some pathologies (Wei et al., 2019). For instance, the onset of schizophrenia has been linked to susceptibility genes coding for several proteins involved in synaptic selection or pruning, such as ERBB4, SLC1A3, RAPGEF4, and CIT28; the last is also involved in bipolar disorder (Karlsgodt et al., 2008). Links with NEUREXIN 1, which is involved in synapse formation and stabilization, have been reported (Cook & Scherer, 2008). Also, various mutations linked to autism are in genes that are involved in synapse formation and stabilization such as NEUROLIGINS 3/4, NEUREXIN 1, and SHANK 3, which code for synaptic adhesion and stability proteins (Bourgeron, 2009; Huguet et al., 2016). Significantly, the long-range connections might be affected differentially by susceptibility mutations that are known to affect synaptogenesis in general (Scott-Van Zeeland et al., 2010). In addition, increased synapse elimination by microglia in schizophrenia patient-derived models of synaptic pruning has been observed (Sellgren et al., 2019). A particular vulnerability of long-range connections might result, for instance, from a very low nucleo-cytoplasmic ratio and/or changes in the long-distance transport of essential cellular components along the axons, which could explain the specificity of the schizophrenic or autistic phenotypes as distinct from the mental retardation expected from global deficits in synaptogenesis.

One way this vulnerability was tested was to compare the dendritic branching of pyramidal neurons in wild-type mice and mutant mice, mentioned above, which lack the beta 2-subunit of the nAChR (Ballesteros-Yáñez et al., 2010). Loss of this subunit prevents the high-affinity binding of the neurotransmitter acetylcholine and, therefore, mice lacking the beta 2-subunit show a characteristic behavioral deficit: their exploratory drive, which is one of the most cognitive aspect of mouse behavior, is reduced, although their navigation abilities, a more automatic activity, are unaffected. Even though establishing a fair analogy between mouse behavior and human psychology may look far-fetched, it has been hypothesized that these mice might possibly be showing an alteration in elementary conscious access (Avale et al., 2011; Koukouli et al., 2017).

These studies on mice may be relevant to a possible effect of chronic nicotine use on long-range connectivity. In humans, diffusion tensor imaging (DTI), which allows the measurement of the location, orientation, and anisotropy of the white-matter tracts in the brain, has shown reduced integrity in the frontal white matter in people who are cocaine dependent or who abuse heroin. The same method has revealed that prenatal and adolescent exposure to tobacco smoke alters the development of the microstructure of the white matter, with increased fractional anisotropy in the right and left frontal regions, and in the genu of the corpus callosum (Changeux & Lou, 2011; Kangiser et al., 2019). These observations suggest that nicotine may also act directly on white matter and there is electrophysiological evidence that supports a direct action of nicotine on axon conduction, possibly at the level of the node of Ranvier (Changeux, 2010). Thus, there is support for an epigenetic control of the global neuronal workspace connectivity by nicotine at the white matter level. This work further implies that drugs of abuse like nicotine may interfere with the functioning of the long-range cortical

connections, to the extent that addicts may lose some conscious self-control of their actions (Changeux & Lou, 2011; Joutsa et al., 2011; Rømer Thomsen et al., 2013). These observations strengthen the still hypothetical conclusion that the epigenesis of long-range connections plays a critical role in elementary conscious access.

2.7 Social and Cultural Imprints

The extension of the postnatal period of development in humans has been essential for the genesis and internalization of culture, as well as for the acquisition and transmission of individual experience. Among the many manifestations of cultural evolution, writing, and reading appear as recent inventions. Writing can be traced back to abstract cave drawings, dated around 30,000 BCE. Clay counting tokens are known from Mesopotamia (9,000 BCE) and the first pictograms from Ur are from around 4,000 BCE. Writing and reading is a recent cultural invention that evolved into distinct sub-systems and puts considerable demands to our cognitive system. The acquisition of reading and writing may be viewed as a typical example of epigenetically led down "cultural circuits."

Historically, the first evidence for specialized writing and reading circuits in the brain was the discovery by the French neurologist Dejerine (1914) of pure alexia, also known as alexia without agraphia. Individuals with pure alexia suffer from severe reading problems, while other language-related skills such as naming, oral repetition, auditory comprehension, or writing are typically intact. Alexia results from cerebral lesions in circumscribed brain regions including the supramarginal and angular gyri. New specialized sets of connections have been selected and consolidated as a consequence of written language learning. More recently, the circuits involved in literacy have been examined using brain imaging. They confirm Dejerine's pioneering insight. These studies (Castro-Caldas et al., 1998) took advantage of behavioral evidence of different phonological processing in illiterate versus literate subjects. During repetition of real words, the literate and illiterate groups performed similarly and activated similar areas of the brain. In contrast, illiterate subjects had more difficulty correctly repeating pseudowords and did not activate the same neural structures as literates. Comparison of positron-emission tomography (PET) scans from illiterate and literate groups showed a considerable shift in activation. For instance, in a pseudowords–words contrast, activation in the literate group was stronger in the right frontal opercular–anterior insular region, left anterior cingulate, left lentiform nucleus and anterior thalamus/hypothalamus compared with the illiterate group (Carreiras et al., 2009; Castro-Caldas et al., 1998). These conclusions have been consolidated and further expanded with functional magnetic resonance imaging (fMRI) scans from illiterate and literate groups (Dehaene et al., 2010). Acquisition of reading and writing may be viewed as an example of epigenetically laid down "cultural circuits" following epigenetic appropriation of fast developing connections around five-to-six years of age. It operates at a time of still very rapid synaptogenesis and persists into adulthood (Dehaene et al., 2010).

Interestingly, occidental alphabetic writing systems recruit circuits which differ in part from those mobilized by the Chinese ideographic systems. In French readers reading French, activations were enhanced in left-hemisphere visual area V1, with the strongest differences between French words and their controls found at the central and horizontal meridian representations. In contrast, Chinese readers reading Chinese showed enhanced activations in intermediate visual areas

V3v/hV4, absent in French participants (Szwed et al., 2014).

Written language learning is one of many social and cultural imprints acquired during the human brain's development. Cognitive development per se, i.e. acquisition of *object* unit and permanence in infants, then *number*, *categorization* and *reasoning* operations in children and adolescents, depend both on brain inner regulation processes and socio-cultural imprints (Houdé, 2000, 2019). Thus, the adult human brain connectivity may be viewed as a complex intertwining of cognitive, social, and cultural circuits epigenetically laid down during development within the framework of a human-specific genetic envelope. Important bridges could then be established between the gene expression level and the highest level of the interaction of the brain with its social and cultural environment.

2.8 Conclusion: The Habitus of the Human Subject

Even if advocated by such a distinguished scholar as Piaget (1976), there is no compelling evidence whatsoever that the culturally acquired phenotypes may sooner or later become genetically transmitted. They have to be learned at each generation from adults to children and epigenetically transmitted from generation to generation, starting even in the womb of the mother, until the adult stage. Teaching reading and writing in five-to-six-year-old children requires elaborate pedagogic strategies, which in a general manner are absent in non-human primates (Premack, 2007).

As a major consequence, the acquisition of skills and practices associated with the symbolic experience and emotional labelling characteristic of rational thinking (science), rules of conduct (ethics), and shared feelings (art) become stably internalized epigenetically in the connectivity of the brain and become exclusive of others. Their "forgetting" is considerably slower than their speed of acquisition through synapse stabilization processes. They contribute to what Bourdieu refers to as the "habitus" of the subject: internalized routine modes of perception, action, and evaluation, which often last a lifetime (Changeux, 2006). Possibly, since the domestication of fire, the grouping of human individuals into stable, autonomous, and geographically dispersed societies has led to the differentiation and divergence of cultures between human groups with separate languages, knowledge backgrounds, techniques and, most of all, systems of beliefs, thus raising cultural fences. These fences, as already noted by Levi Strauss (1952), are such that, because of its different culture, "the other" may no longer be regarded as a human being. A "dehumanization" process frequently accompanies the diversification of cultures (Changeux, 2018, 2019). Most often, a widespread "ethnocentrism" of the subject takes place, where he/she considers his/her own culture as "The only One Culture" existing on earth. Such cultural habitus acquired in the child and "printed" in its brain remains stable for years or even decades. It manifests itself, for instance, in the accent in spoken language. It can only be fully renewed with the re-learning step that takes place with the next generation (Changeux, 2018, 2019).

Finally, social and cultural evolution is associated with variable synaptic efficacy and the establishment of extracerebral memories in the form of spoken, written, and pictorial material, with a time range of 100 milliseconds to thousands of years. Spoken and written language and, perhaps even more significantly, artistic activity are seminal innovations that distinguish humans from other primates; they drove the development of modern civilization and have probably also been central to the expansion of human mental capacities. More

important for my thesis, language, writing, and cognitive development in general (about objects, numbers, categorizations, and reasonings or decision-makings) rely on epigenetic cultural transmission framed within a robust human-specific genetic envelope. The huge postnatal increase in the size of the human brain – the adult brain weights five times that of the newborn infant and about 50 percent of the adult brain's connections develop after birth (Bourgeois, 1997; Huttenlocher & Dabholkar, 1997) – offers the developing brain the opportunity for intense self-regulations and social and cultural interactions.

From a neuroscientific perspective, ethical and social norms may therefore be conceived as spatiotemporal patterns of neuronal activity that can be mobilized within the "conscious neuronal workspace" and stored as long-term traces in brain memory (Changeux, 2018; Evers & Changeux, 2016; Farisco et al., 2018). In more general terms, the prefrontal region of the cerebral cortex acts as a "temporal buffer" between past events and future actions and contributes to decision-making within the context of the individual's history. In such a neuronal workspace, "neurally encoded rules" associate a context with a specific behavioral response in a top-down manner. This is referred to as cognitive control (Houdé, 2019) and coordinates thoughts or actions in relation to internal goals. Brain imaging fMRI revealed a hierarchical cascade of executive processes, which are implemented in distinct regions, from the posterior premotor to rostral lateral prefrontal cortex, typically Brodman's area 46 (Houdé & Tzourio-Mazoyer, 2003; Houdé et al., 2000, 2011; Koechlin, 2016). Behavioral rules are sorted at these nested levels of information processing, the highest-level ones controlling the underlying ones closer to the senses. This scheme may be extended assuming that ethical or social norms control rules and part of the "concurrent behavioral strategies" in decision-making (Evers & Changeux, 2016). From birth on, and possibly even prenatally, the baby is exposed to a social and cultural environment. During its development, an epigenetic selection of neuronal networks accompanies the acquisition of the ethical rules of the social community to which the child and his/her family belong. These ethical rules are often linked with symbolic representations of the cultural community, and the acquisition of such ethical rules and symbolic systems has been compared to language acquisition. Yet, to justify such cultural differences, written language, holy books, moral rules, sexual practices, food, or clothing rituals among others are evoked as irreducible transcendentals even more fundamental than political or economical differences. The universe of cultural differences and thus of potential conflicts is immense. How to face this dramatic situation? How to overcome the cultural epigenetic fences that plague our planet?

One possibility to consider is to make an effort to gain some distance with respect to the emotions mobilized by what is believed to be a "transcendental" act of faith. This might be achieved, at the individual level, through the understanding that such belief is simply an acquired cultural trait invented and epigenetically perpetuated by the brains of humans in society to consolidate their social bond. Following Durkheim, such social facts would possess an objective reality that could be "studied like a physicist studies the physical world," taking into account the fact that they are produced and propagated in a steadily evolving socio-historic context. Within this context, the possibility of developing in the brain new ethical rules of "good life with and for the other" in just institutions (Changeux & Ricoeur, 2000) or "epigenetic proaction" has been proposed (Evers & Changeux, 2016). Another possibility, advocated by Levi-Strauss, is to look for a

"coalition of cultures." Such a coalition would favor the evolution of cultures supporting progress and cooperation rather than exclusion, resulting in the breakdown of the cultural fences.

Last, a possibility – seemingly easier to achieve – is to be found in education. Cultural differences are infused into the child brain already at birth (possibly even before for language), in the family environment and consolidated by the schooling system. Secular education is a system of public education, where conspicuous symbolic (religious or philosophical) systems as well as the biased unilateral presentation of the diversity of symbolic systems have been banned. Still it has been recognized as one of the best systems in the world to learn and practice tolerance and respect for cultural differences (Changeux, 2018; Evers & Changeux 2016).

In conclusion, we are neurobiologically predisposed toward specific values, such as self-interest, empathy, sociality and so on, and our brain structures develop in response to ethical and social norms in our cultural and social context. Given the long-lasting plasticity of our brains and the underlying synaptic epigenesis mechanisms involved, we may influence, both biologically and culturally, how the brain responds to and constructs new norms with the aim to account for a harmonious future of humanity in our fast-evolving planet.

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3 Mapping the Human Brain from the Prenatal Period to Infancy Using 3D Magnetic Resonance Imaging

Cortical Folding and Early Grey and White Maturation Processes

Arnaud Cachia, Jean-François Mangin, and Jessica Dubois

3.1 Introduction

Human brain development is a complex and dynamic process that begins during the first weeks of pregnancy and lasts until early adulthood. This chapter will focus on the developmental window from the prenatal period to infancy, probably the most dynamic period across the entire lifespan. The availability of non-invasive three-dimensional Magnetic Resonance Imaging (MRI) methodologies has changed the paradigm and allows investigations of the living human brain structure – for example, micro- and macrostructural features of cortical and subcortical regions and their connections, including cortical sulcation/gyrification, area, and thickness, as well as white matter microstructure and connectivity, see Boxes 1–3 (Sections 3.6.1–3.6.3) – beginning *in utero*. Because of its relative safety, MRI is well-adapted to study individuals at multiple time points and to longitudinally follow the changes in brain structure and function that underlie the early stages of cognitive development.

Before the advent of brain imaging tools, structural brain changes were inferred from *post-mortem* data with major concerns about their generalizability due to the questionable good health of the studied individuals with very young death. On the other hand, despite the advantages of *in vivo* MRI, MRI cannot measure structural changes at the cellular or molecular levels, and the physiological interpretation of MRI signal is not straightforward nor univocal. A detailed description of early developmental mechanisms (e.g., formation of the neural tube, neuronal migration, and differentiation), as well as the early brain organization in transitory compartments such as the subplate can be found in complementary reviews (Kostovic & Judas, 2015; Stiles & Jernigan, 2010).

In addition, MRI data acquired on the early stage of brain development are very noisy due to the acquisition constraints (e.g., fetus movements, short acquisition duration), and difficult to analyze due to the low spatial resolution and the age-dependent tissue contrast and structure size. The comparison of anatomical brain measures derived from MRI across ages should therefore remain cautious. However, despite these limitations, the

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spatial and temporal patterns of developmental changes observed in recent MRI studies reflect patterns that were observed *post-mortem*, demonstrating the validity and compatibility of these methods (Dehaene-Lambertz & Spelke, 2015; Dubois et al., 2015)

We here review how the cortex grows and gets convoluted, the microstructural maturation of the gray and white matter, and their cognitive correlates in normal condition (see Anderson et al., 2011; Miller et al., 2016, in pathological conditions).

3.2 The Early Development of the Brain Cortex

3.2.1 Cortical Volume

The last weeks of pregnancy and the first postnatal months are marked by an intense increase in cortical volume (Dubois & Dehaene-Lambertz, 2015) which progressively slows down after two years of age until adolescence. Changes in cortical volume are driven by changes in cortical thickness (CT) and cortical surface area (CSA), two complementary macrostructural features of the cortex anatomy with distinct genetic (Panizzon et al., 2009; Raznahan et al., 2011) and developmental (Raznahan et al., 2011) mechanisms. The classical "radial unit hypothesis" (Rakic, 1988, 2000) assumes that the CSA of a given cortical area essentially reflects the number of cortical columns, while CT essentially reflects the number and size of cells within a column and packing density, as well as the number of connections and the extent of their myelination (Eickhoff et al., 2005).

During fetal life, the total cortical volume increases from 10 cm^3 at 18 weeks of gestational age (wGA), to 30 cm^3 at 27 wGA, and 150 cm^3 at 39 wGA (Andescavage et al., 2017; Makropoulos et al., 2016), reaching 200 cm^3 one month after birth and 600 cm^3 at two years old (Knickmeyer et al., 2008) (Figure 3.1a). The total CSA of the fetus brain at 27 wGA is around 150 cm^2 (Makropoulos et al., 2016), reaching 600–800 cm^2 one month after birth, 1,300 cm^2 at five months, and 2,000 cm^2 at twenty-four months (Dubois et al., 2019; Lyall et al., 2015).

The cortical volume increase is more important during the first postnatal year (~100 percent) than during the second year (~20 percent) (Gilmore et al., 2012). Developmental rates differ among brain regions, with important cortical volume increases in parietal and occipital regions in utero (Rajagopalan et al., 2011) and in association cortices, particularly in the frontal and parietal lobes, after birth (Gilmore et al., 2012). During the first two years after birth, brain growth is mainly due to gray matter development (Gilmore et al., 2007; Knickmeyer et al., 2008), which is no more the case later (Matsuzawa et al., 2001).

3.2.2 Cortical Morphology

Parallel to volumetric changes, early brain development is characterized by dramatic changes in cortex morphology due to the cortical folding process that begins from ten weeks of fetal life (Feess-Higgins & Larroche, 1987; Welker, 1988) (Figure 3.1a and Box 2, Section 3.6.2). During the third trimester of pregnancy, the cerebral cortex changes from a relatively smooth, lissencephalic, surface to a complex folded structure that closely resembles the morphology of the adult cortex (Figure 3.1a). The precise mechanism underlying cortical folding is still unknown. However, several factors likely contribute to the prenatal processes that influence the shape of the folded cerebral cortex, including cortical growth, apoptosis (i.e., programmed cell death), differential expansion of superior and inferior cortical layers, differential growth of the cortical mantle relatively to the underlying white

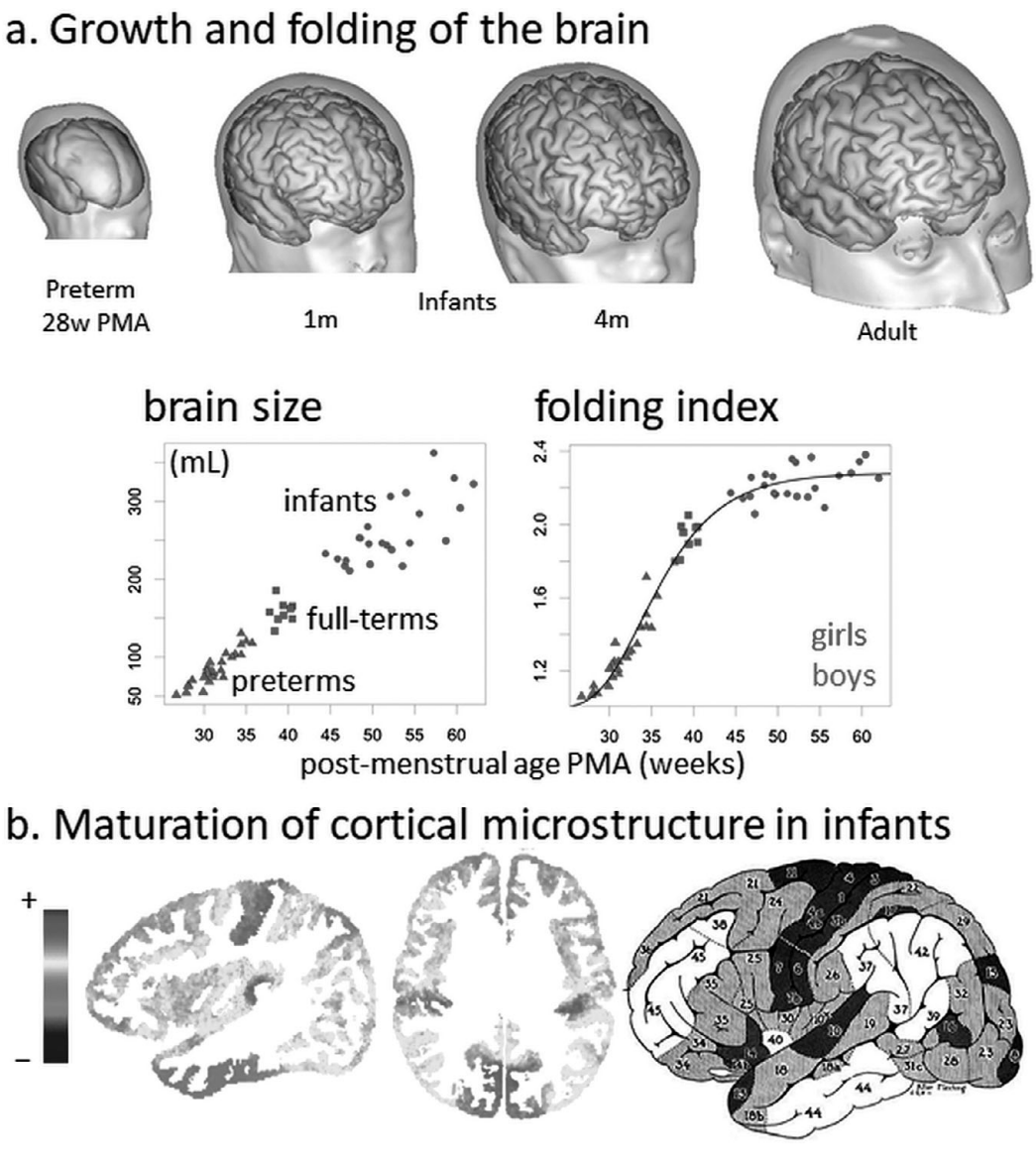


Figure 3.1 Structural changes of the brain during development. (a) The brain shows intense growth and folding during the last weeks of pregnancy and the first months of infancy, as demonstrated with anatomical MRI. Illustrations are provided for a preterm newborn at twenty-eight weeks of post-menstrual age (PMA), infants (at one and four months after birth), and an adult. Measures of brain size and folding index as a function of age provide a quantitative illustration of these intense changes (adapted from Dubois et al. (2019)). (b) The cortex also shows dramatic changes in microstructure complexity and maturation. A recent multi-parametric MRI approach highlighted strong differences between cortical regions in infants between one and five months of age. Colored maps adapted from Lebenberg et al. (2019), in agreement with post-mortem map of subcortical myelination by Flechsig (1920) adapted from Von Bonin (1950)

A black-and-white version of this figure will appear in some formats. For the color version, please refer to the plate section.

matter (Tallinen et al., 2016), transitory compartments such as the subplate (Rana et al., 2019), differential neuropil developments (Llinares-Benadero & Borrell, 2019; Mangin et al., 2019), differential tangential expansion induced by a genetics-based protomap, and/or structural connectivity through axonal tension forces (see Borrell, 2018; Foubet et al., 2019; Kroenke & Bayly, 2018; Zilles et al., 2013, for recent reviews on the phylogenetic, cellular, and mechanical factors of the cortical folding process). The folding process might enable an increase in the cortical surface area while maintaining the volume of the axons required for interconnecting the areas and thus relatively short distances and reasonable communication times between distant brain regions (Klyachko & Stevens, 2003). Macroscopic (morphological/volumetric) and microscopic (cellular) features of the cortex are therefore intrinsically interrelated. For instance, it has been shown that the cortical ribbon is thicker in the gyral areas and thinner in the sulcal areas and neurons located in the deep layers of gyri are squeezed from the sides and appear elongated while neurons that reside in the deep layers of sulci are stretched and look flattened (Hilgetag & Barbas, 2006, 2009).

Dedicated MRI tools and morphometric analyses have enabled us to map in detail the developing cortical surface and growth patterns in fetuses as young as 20 wGA (Habas et al., 2012). These in vivo studies confirm earlier post-mortem observations (Chi et al., 1977) and show a precise timing: stable primary folds appear around 20 wGA, secondary folds around 32 wGA, and highly variable tertiary folds around term (Chi et al., 1977). The gyrification (the apparition of the gyri, the "mountains" of the cortical relief) and the sulcation (the apparition of the sulci, the "valleys" of the cortical relief) become manifest after 24 wGA (Rajagopalan et al., 2011), and greatly heightens during the last weeks before birth (Dubois et al., 2008b, 2019). Although some variability is observed among individuals, the regional pattern is consistent over the brain surface: sulcation starts in the central region and proceeds first toward the parietal, temporal, and occipital lobes, and second toward the frontal lobe (Dubois et al., 2008b; Ruoss et al., 2001). The heritability of the cortical folding is estimated between 0.2 and 0.5 (Le Guen et al., 2018), supporting a major role of early environmental factors like alcohol exposure (De Guio et al., 2014), intrauterine growth restriction, or twin pregnancy (Dubois et al., 2008a) in the cortical folding process. The morphology of the cortical folding at birth can then be considered a retrospective marker of fetal brain development. It can also be used as a prognostic marker of later functional development. Indeed, cortical sulcation at birth in preterms has been shown to predict infants' neurobehavioral development several weeks later (Dubois et al., 2008a; Kersbergen et al., 2016).

The development of the volumetric and morphological features of the brain, although governed by specific mechanisms, are not independent. The cortical folding index scales uniformly across species and individuals as a function of the product of CSA and the square root of CT (Mota & Herculano-Houzel, 2015). As early as 23 wGA, CSA and total brain volume were also found to be related by a scaling law whose exponent predicts later neurodevelopmental impairment in preterms (Kapellou et al., 2006).

At birth, the cortical surface area is three times smaller than in adults, but the cortex is roughly similarly folded, and the most variable regions among individuals are the same across newborns and adults (Hill et al., 2010). Unlike quantitative features of cortical anatomy such as CSA or CT which can take decades to attain the levels observed in adulthood (Giedd & Rapoport, 2010; Li et al., 2014; Raznahan et al., 2011), the qualitative features of the cortex anatomy, such as the sulcal patterns

(Cachia et al., 2016; Tissier et al., 2018) or the incomplete hippocampal inversion (Bajic et al., 2008; Cury et al., 2015) are determined in utero and are stable after birth. The analysis of such trait features of the brain can thus provide information on the prenatal constraints imposed by the structure of some specific brain regions on later cognitive development.

3.2.3 Study of the Fetal Foundation of Cognition Using the Sulcal Patterns

Several studies have reported that subtle variations of the in utero environment, as indexed by birth weight, are accompanied by differences in postnatal cognitive abilities (Raznahan et al., 2012; Shenkin et al., 2004; Walhovd et al., 2012). In addition to such a global proxy measure of "uterine optimality" (Raznahan et al., 2012), analysis of the sulcal patterns can provide information on the prenatal constraints imposed by the structure of some specific brain regions on later cognitive development.

Several studies in typically developed participants reported long-term influence of cortical sulcation at birth on cognition several years and decades later (for a review in impaired cognition, see Mangin et al., 2010). For instance, a critical region of the cognitive control is the dorsal anterior cingulate cortex (ACC) (Petersen & Posner, 2012) which presents two qualitatively distinct sulcal patterns: a "single" type when only the cingulate sulcus is present and a "double parallel" type when a paracingulate sulcus runs parallel to the cingulate sulcus. An asymmetrical sulcal pattern of the ACC (i.e., different sulcal pattern in left and right hemispheres) was found to be associated with higher cognitive control in children at age five (Cachia et al., 2014), and four years later (Borst et al., 2014), as well as in adults (Fornito et al., 2004; Tissier et al., 2018) (Figure 3.2). A similar effect was found in children and adults for the inferior frontal cortex (IFC), another key region of cognitive control (Tissier et al., 2018). These early neurodevelopmental constraints on later cognitive efficacy are not fixed nor deterministic. Indeed, only a part (15–20 percent) of the cognitive variability is explained by the sulcal pattern variability (Borst et al., 2014; Cachia et al., 2014; Tissier et al., 2018). In addition, different environmental backgrounds, either after birth, such as bilingualism (Cachia et al., 2017; Del Maschio et al., 2018), or before birth, such as twin pregnancy (Amiez et al., 2018), can modulate the effect of the sulcal pattern on cognition. Sulcal studies also revealed that cognitive abilities requiring intensive learning and training, such as numeracy or literacy, can also be traced back to early stages of brain development. Indeed, the pattern (continuous or interrupted sulcus) of the posterior part of the left lateral occipito-temporal sulcus (OTS) hosting the visual word form area (VWFA) predicts reading skills in ten-years-old children (Borst et al., 2016) and in adults (Cachia et al., 2018) (Figure 3.2). The presence or absence of branches sectioning the horizontal branch of the intra-parietal sulcus (HIPS), a key region for processing numbers, is related to individual differences in symbolic number comparison and math fluency abilities in children and adults.[1]

3.2.4 Cortical Microstructure

These macro-structural changes, with increasing cortical folding and thickness with age, are probably the visible markers of the microstructural evolution occurring in the cortical plate over preterm and early post-term, marked by synaptic outburst and pruning, modifications

[1] Roell, M., Cachia, A., Matejko, A., Houdé, O., Ansari, D., & Borst, G. (Submitted). Sulcation of the intraparietal sulcus is related to childrens symbolic but not non-symbolic number skills.

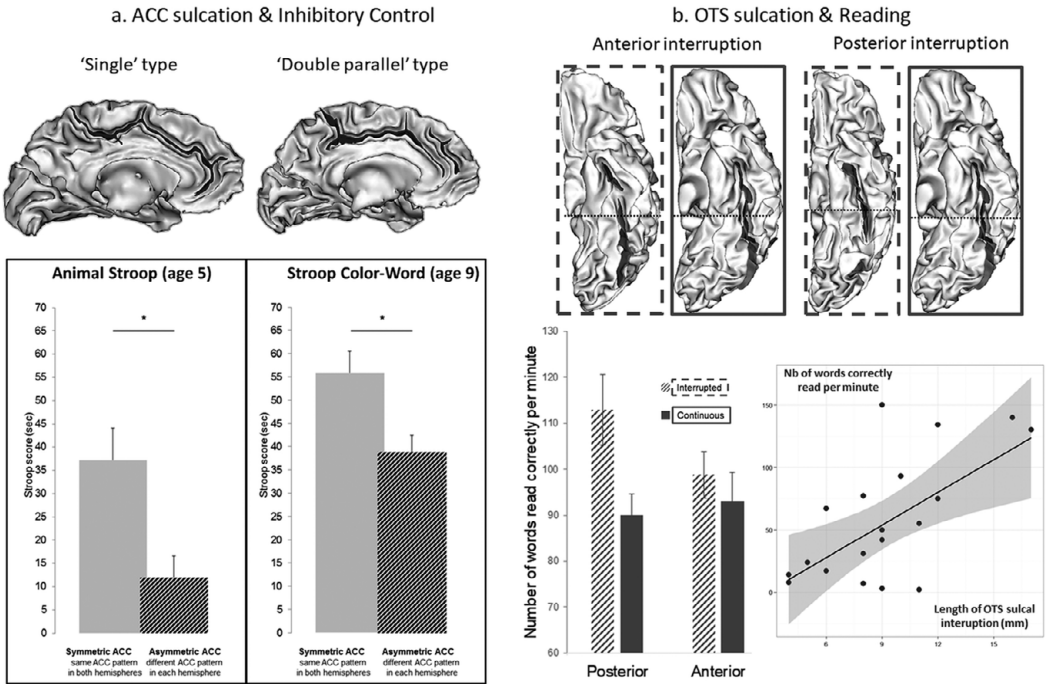


Figure 3.2 Cortical sulcation and cognitive efficiency. (a) Effects of the sulcation of the anterior cingulate cortex (ACC) on inhibitory control efficiency. Upper panel: the ACC can have two types of sulcal patterns: "single" type when only the cingulate sulcus was present and "double parallel" type when a paracingulate sulcus ran parallel to the cingulate sulcus (sulci are depicted in blue on the gray/white interface). Lower panel: Mean Stroop interference scores at age 5 (on the Animal Stroop task) and at age 9 (on the Color-Word Stroop task) in children with symmetrical (single or double parallel type in both hemispheres) and asymmetrical (single type in the right hemisphere and double type in the left hemisphere or vice versa) ACC sulcal patterns. Children with asymmetrical ACC sulcal patterns have better inhibitory control efficiency than children with symmetrical ACC sulcal patterns. Adapted from Borst et al. (2014) and Cachia et al. (2014). (b) Effect of Occipito-Temporal Sulcus (OTS) sulcation on reading ability. Upper panel: the left posterior OTS hosting the visual word form area (VWFA) can have two types of sulcal patterns: "continuous" type or "interrupted" type in case of a sulcal interruption located posteriorly, below the back dashed line. Lower panel: Number of words reads correctly in children or adult participants with continuous (in plain gray) or interrupted (in hatched gray) left OTS. Participants with interrupted OTS have better reading efficiency than participants with continuous OTS. The number of words read per minute is positively correlated with the length of the sulcal interruption. Adapted from Borst et al. (2016) and Cachia et al. (2018)
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in dendritic branching and fiber myelination. For instance, histological studies showed that cortical synaptogenesis starts from 16 wGA, and the peak in synapse density depends on the brain region (Huttenlocher & Dabholkar, 1997; Kwan et al., 2012).

In recent years, several MRI methods have been used to provide quantitative information

on these mechanisms in vivo (Box 3, Section 3.6.3). Diffusion MRI, notably diffusion tensor imaging, or DTI (Huppi & Dubois, 2006), has shown the different brain layers (cortical plate, subplate, inner layer) in post-mortem fetuses from 13 wGA (Huang et al., 2013, 2009) and in vivo preterm newborns from 25 wPMA (Maas et al., 2004). In the cortical plate, a complex age-related evolution is observed for the diffusion properties of water molecules (Dudink et al., 2010; McKinstry et al., 2002). This tissue first shows some anisotropy from 26 wGA/PMA with a radial orientation of the main tensor direction which might rely on the early radial deployment of glial fibers and apical dendrites of pyramidal neurons (Ball et al., 2013; Ouyang et al., 2019). This transient radial alignment was quantified by a radiality index that measures the directional coherence between the diffusion tensor and the cortical surface (Eaton-Rosen et al., 2017). Thereafter the cortical plate becomes isotropic from around 36 wGA/PMA with the elongation and complex branching of neuronal connections (basal dendrites for the pyramidal neurons and thalamo-cortical afferents), and the decrease in cortical anisotropy seems to stop at around 38 wPMA (Batalle et al., 2019). Besides, some reported an increase in DTI mean diffusivity between 26 and 32 wPMA (McKinstry et al., 2002), whereas most others have described a steady decrease with age (Batalle et al., 2019; Ouyang et al., 2019). This suggested competing mechanisms, such as a decrease in neuronal density associated with programmed cell death versus an increase in glial cells, addition of neuropils between the neuronal somas, decreasing water content, etc. To decipher between these complex microstructural changes, recent studies have used more elaborate diffusion models (e.g. Neurite Orientation Dispersion and Density Imaging, NODDI) (Batalle et al., 2019; Eaton-Rosen et al., 2015). Before 38wPMA, the neurite density index (NDI) decreases and the orientation dispersion index (ODI) increases, consistently with a predominant increase in dendritic arborization and neurite growth. After this age, ODI plateaus and NDI increases in primary sensori-motor regions, suggesting that the 38–47 wPMA period might be dominated by increasing cellular and organelle density. Thus these microstructural changes are not linear, at least for some regions, and they are not uniform over the brain (Ball et al., 2013; Deipolyi et al., 2005). In particular, cortical sulci might show a more complex microstructure than gyri early on (Ball et al., 2013).

Diffusion parameters are also sensitive to the regional heterogeneity in cortical development. The occipital lobe shows the most rapid decrease in radiality index, and the frontal and temporal the least ones (Eaton-Rosen et al., 2017). Similarly, functionally distinct regions show different patterns of anistropy decrease from 20 to 35 wGA (Yu et al., 2017), which highlights the asynchronous rate of maturation across cortical areas. DTI parameters further revealed advanced cortical maturation of the primary auditory cortex by 28 wPMA, while rapid changes take place in the non-primary cortex of Heschl's gyrus between 26 and 42 wPMA (Monson et al., 2018). In a recent exploratory study, sharper changes in cortical microstructure were related to a more rapid increase in cerebral blood flow in preterms from 32 to 45 wPMA (Ouyang et al., 2017b).

During infancy, the evolution of cortical microstructure is complex. The growth of connections between neurons is first intense and exuberant through synaptogenesis and the extension of dendritic arborization. Then this phase is followed by an elimination of useless connections through a pruning mechanism, to select and maintain only functionally efficient connections. This occurs over different age periods for different cortical regions depending on the functions and the environmental

stimulations. Accompanying this process, the dendritic and axonal fibers get myelinated, mostly during the early post-term period. Post-mortem studies showed that brain regions have different maturational trajectories relatively to the myelination of sub-cortical white matter fibers (Flechsig, 1920). This mechanism can be explored *in vivo* with MRI by taking advantages of changes in T1 and T2 signals (two complementary standard macroscopic measures depending on different features of the brain anatomy at cellular level) induced by modifications in water and iron contents (Ouyang et al., 2019). Differences in myelination across cortical regions get stronger from 36 to 44 wPMA, as observed in premature infants with the T1w/T2w ratio (Bozek et al., 2018). The asynchrony of maturation across primary and associative regions was also recovered over the infant language network during the first postnatal months based on T2w images (Leroy et al., 2011), and during the second year based on T1w images (Travis et al., 2014). Multiparametric MRI approaches combining DTI, relaxometry (e.g., quantitative T1), or multi-compartment approaches (e.g., myelin water fraction) have also highlighted differences in the microstructural and maturational properties of cortical regions in preterms at term equivalent age (Friedrichs-Maeder et al., 2017), in infants (Lebenberg et al., 2019) (Figure 3.1b), and in toddlers (Deoni et al., 2015). All these changes observed with distinct MRI measurements probably reflect a complex interplay between several mechanisms, such as the development of dendritic arborization, the proliferation of glial cells, and/or the myelination of intracortical fibers.

3.2.5 Maturation of Central Grey Matter Nuclei

Marked microstructural changes are also observed in central gray nuclei throughout development, with an age-related decrease in DTI diffusivities and an increase in anisotropy over the preterm period and infancy (Mukherjee et al., 2001; Neil et al., 1998; Qiu et al., 2013), suggesting intense membrane proliferation and fiber myelination. In parallel and simultaneously with white matter regions, T1 decreases gradually over central gray nuclei (Schneider et al., 2016), as T2 does (Bultmann, Spineli, Hartmann, & Lanfermann, 2018). The developing microstructural properties are better characterized by comparing multiple parameters (e.g., DTI, NODDI, T1, magnetization transfer ratio, fraction of water related to myelin). This enables us to disentangle between maturational mechanisms (e.g., concentration of myelin-associated macromolecules, water content) (Melbourne et al., 2016; Nossin-Manor et al., 2013). Much like the cortical regions, the maturational patterns differ across nuclei, notably across thalamic substructures (Poh et al., 2015).

3.3 The Early Development of White Matter

In interaction with the development of cortical regions, intense and intermingled processes of growth and maturation are occurring within the brain white matter from the preterm period to infancy (Dubois et al., 2014, 2015, 2016a). All major long-distance fibers are observed by term birth (Figure 3.3a), while short-distance fibers (e.g. U-fibers) mainly develop during the first post-natal year. All these anatomical connections further refine through several complementary mechanisms. After an exuberant growth and proliferation, useless and redundant connections are pruned during childhood, whereas the ongoing myelination process stabilizes the functionally relevant ones and increases the efficiency of the information transfer between distant brain regions. Whereas the number of neurons and microglia

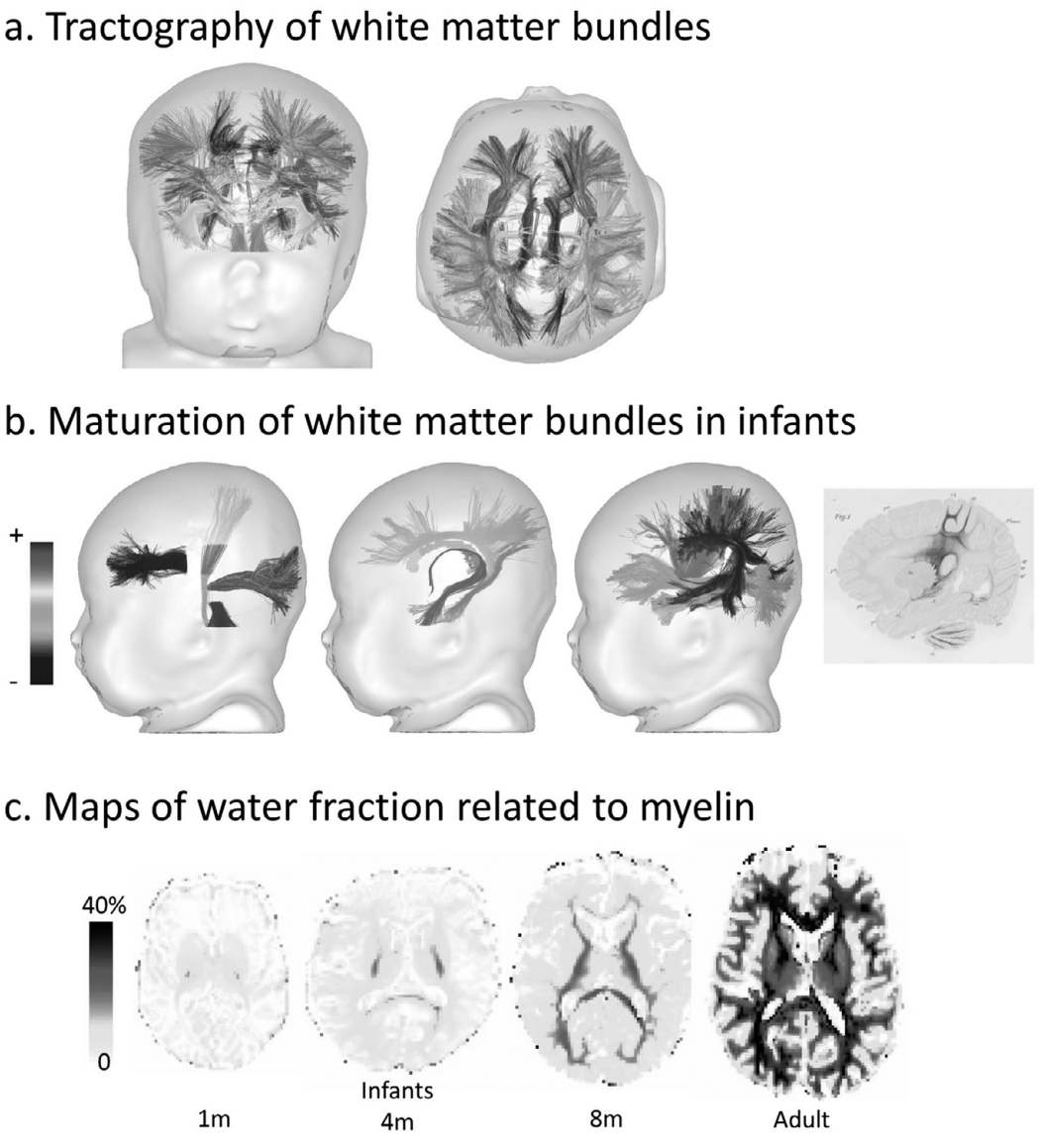


Figure 3.3 Structural changes of the white matter during development. (a) The brain shows an early organization in networks, with white matter bundles connecting distant and close regions, as illustrated here with diffusion MRI and tractography reconstructions in a one-month old infant. (b) The white matter also shows intense maturation after birth through the myelination process (Yakovlev & Lecours, 1967). A recent multi-parametric MRI approach confirmed strong differences between bundles in infants between one and five months of age (adapted from Dubois et al. (2015) and Kulikova et al. (2015) in agreement with a post-mortem map of subcortical myelination by Flechsig (1920) adapted from Von Bonin (1950)). (c) Different MRI techniques inform on myelination, such as the one computing the fraction of water related to myelin. These maps highlight the progression of myelination from central regions to the periphery. Adapted from Kulikova et al. (2016)
A black-and-white version of this figure will appear in some formats. For the color version, please refer to the plate section.

remains almost the same postnatally, the number of oligodendrocytes and astrocytes drastically increases in the white matter during the first 3 years, attaining two-thirds of the corresponding numbers in adults (Sigaard et al., 2016). The inhibitory role of oligodendrocytes and myelin on neuritic growth may partly explain the weak plasticity of the adult brain (Ng et al., 1996). In addition to the early brain organization in specific networks, a major developmental characteristic is the asynchronous progression of maturation across brain regions: for instance, sensory regions develop early on and quickly, whereas associative regions are slowly developing until the end of adolescence.

3.3.1 Developing Architecture of White Matter Bundles

Recent post-mortem and in vivo studies using diffusion MRI combined with physical models (e.g., diffusion tensor imaging DTI, high angular resolution diffusion imaging HARDI) and sophisticated tractography tools (cf. Box 3, Section 3.6.3) have confirmed and extended the knowledge on bundles growth and organization which previously relied on histological dissections. Nevertheless, several issues are related to these techniques in the developing brain (Dubois et al., 2016a). For instance, fetal and neonatal MRI is limited by motion-related constraints and spatial resolution issues (Dubois et al., 2014). Besides, the biological interpretation is not straightforward: not only do the axons participate in the depicted anisotropy of water diffusion, but also the radial glial fibers and the penetrating blood vessels, particularly during the preterm period (Xu et al., 2014).

The development of fiber systems proceeds in a sequential way (projection, callosal, and association) (Vasung et al., 2017), and transient fetal patterns of connectivity are observed before the formation of stable connections (Dubois et al., 2015; Kostovic & Jovanov-Milosevic, 2006; Vasung et al., 2010). During the early fetal period (9–15 weeks post-conception wPC), major fiber pathways are growing and pathfinding within the intermediate zone. Some afferent (e.g., thalamo-cortical) and efferent (cortico-subcortical) fibers have already been observed, as well as several limbic bundles (e.g., fornix, cingulum) (Huang et al., 2006; Vasung et al., 2010). Before penetrating into the cortical plate, thalamo-cortical fibers make connections with neurons of the subplate zone (Kostovic & Rakic, 1990). During the midfetal period (15–23 wPC), major efferent fibers penetrate their targets in sub-cortical structures, and cortico-cortical callosal fibers and some associative connections (e.g., uncinate, inferior longitudinal, fronto-occipital fasciculi) begin to emerge (Huang et al., 2009; Judas et al., 2005; Vasung et al., 2017). In fetuses in utero, the pyramidal tract, the splenium, and genu of the corpus callosum are observed from 18 weeks of gestational age (wGA) (Bui et al., 2006; Kasprian et al., 2008; Pontabry et al., 2013), while the uncinate and inferior fronto-occipital fascicles are depicted from 20 wGA (Mitter et al., 2015).

During the early preterm period (24–28 wPC), afferent fibers that were waiting in the subplate start making connections within neurons of the cortical plate (of the future cortical layer IV), leading to the establishment of the permanent connectivity while the transient fetal circuitry still exists within the subplate (Kostovic & Judas, 2006; Vasung et al., 2016). Limbic cortico-cortical connections are well developed in the cingulate, entorhinal and hippocampal cortices (Kostovic et al., 1993). Some associative (e.g. inferior longitudinal fasciculus) and limbic bundles (cingulum, fornix) can be visualized in 3D in fetuses in utero (Mitter et al., 2015). During the late preterm

period (29–34 wPC), long associative and commissural bundles are quickly developing, likely originating from pyramidal neurons of cortical layer III (Schwartz & Goldman-Rakic, 1991), and these bundles can be identified in preterm newborns (Dudink et al., 2007; Partridge et al., 2004). Short cortico-cortical fibers further grow and enter the cortex through the remnant subplate (Kostovic et al., 2014b; Takahashi et al., 2012).

At term birth, elaborate connectivity is established, allowing infants to acquire functional abilities. Notably, in infants born preterm, cognitive performances at two years are related with thalamocortical connectivity at term-equivalent age (Ball et al., 2015). U-fibers further develop (Kostovic et al., 2014a), and cortico-cortical connectivity is reorganized by several processes including the pruning of callosal fibers (LaMantia & Rakic, 1990) which probably extends until the end of the first post-natal year (Kostovic & Rakic, 1990; Kostovic et al., 2014a; Petanjek et al., 2011). During early infancy, all major white matter bundles and short-range connections are identified despite their weak maturation: commissural bundles of the corpus callosum (genu, body and splenium), projection bundles (cortico-spinal and spino-thalamic tracts, optic radiations, anterior limb of the internal capsule), limbic bundles (fornix and cingulum), and associative bundles (external capsule, uncinate, arcuate, superior, and inferior longitudinal fascicles) (Dubois et al., 2006; Kulikova et al., 2015). The trajectory and morphology of these bundles remain stable between birth and two years of age (Geng et al., 2012).

In recent years, the developing architecture of anatomical networks has also been detailed based on connectivity matrices (Keunen et al., 2017b), by measuring the degree of connections for all pairs of brain regions. Already at thirty weeks of post-menstrual age (wPMA), the structural connectome demonstrates a small-world modular organization like in the adult brain (van den Heuvel et al., 2015), and cortical hubs are highly connected to form a "rich club" (Ball et al., 2014). This topology further refines with age (van den Heuvel et al., 2015), becoming more clustered around term (Brown et al., 2014) and showing an increase in global efficiency and integration, and a decrease in segregation during the first two post-natal years (Yap et al., 2011).

3.3.2 White Matter Maturation

Concurrently and subsequently to the development of connections, the white matter bundles progressively mature and become functionally efficient through the myelination process. This includes several steps, and proceeds from the neuron body to the periphery (McCart & Henry, 1994). It occurs in the human brain from the second part of pregnancy to the end of adolescence, with a peak during the first post-natal year (Baumann & Pham-Dinh, 2001; Van der Knaap & Valk, 1995a, 1995b). It progresses at different ages and rates depending on the regions, bundles, and networks (Baumann & Pham-Dinh, 2001; Brody et al., 1987; Flechsig, 1920; Gilles et al., 1983; Kinney et al., 1988; Yakovlev, 1962; Yakovlev & Lecours, 1967), and follows a caudo-rostral gradient, from the center to the periphery, occurring earlier and faster in sensory pathways (somatosensory, vision, audition) than in motor ones, and in projection fibers than in associative ones (Figure 3.3b).

The myelination process can be quantified through several MRI parameters (Dubois et al., 2014, 2016a), which show intense changes after term birth and during the first post-natal months, and differences in maturation across white matter bundles (Dubois et al., 2008c, 2016b; Kulikova et al., 2015). With myelination, the water content decreases, thus the proton density decreases, while the